Software Testing and Quality Evaluation

Emergency Resource Coordination Platform

SE3110 - Quality Management in Software Engineering | Assignment 2

This report evaluates the existing ASP.NET Core, PostgreSQL, React, Flutter and agentic subsystem developed for the SEF assignment. The verified aggregate is 270 passing automated cases, a passing local load scenario and a passing migration CLI check. Two security defects and the migration defect were corrected and retested. All current suites and the load scenario were rerun on 6 October. Counts reconcile automatically with the raw reports; no tests were skipped. The evidence supports the tested workflows; it does not establish production readiness.

Member	Student ID	SEF module
Jathurshy.S	IT24103355	Incident Management and Coordinator/Planner Agent
M.Thanusikan	IT24103609	Resource and Inventory Management and Situation Analysis Agent
Denesh N	IT24102826	Dispatch and Task Assignment and Resource Allocation Agent
Deneshkar. P	IT24103828	Volunteer/Responder Management and Validation/Safety Agent

Executed area	Passed	Failed	Skipped
Backend including PostgreSQL and migrations	95	0	0
React components and services	47	0	0
Flutter units and widgets	91	0	0
Python deterministic agents	20	0	0
Live HTTP workflow and security	17	0	0
k6 load scenario	1	0	0
Migration CLI check	1	0	0

Submission module confirmed by the repository owner: SE3110 - Quality Management in Software Engineering. The supplied brief internally refers to SE3090 and the SEF integrated system; that source wording is retained for traceability. Its stated deadline is 5 October 2026. Later evidence dates are retained accurately; no extension or CourseWeb upload is claimed.

Test plan and environment

Objectives. Verify important implemented behaviour, demonstrate cross-component execution, measure a bounded performance workload, check authorization and safety controls, and preserve traceable defect/retest evidence. The same ERCP source is used throughout.

Scope. Incident reporting and planning; resources, inventory and shelters; dispatch and approval; volunteers and shifts; authentication; React UI states; Flutter models/widgets; deterministic agent contracts; PostgreSQL uniqueness, foreign keys, cascade deletion and rollback.

Area and tools	Reason for selection
xUnit, WebApplicationFactory, EF Core and PostgreSQL	Existing .NET framework; direct business/controller tests plus real HTTP authorization and actual relational constraints.
Vitest and React Testing Library	Existing setup; component rendering, interactions, route permissions and mocked API failure states.
flutter_test	Existing unit/widget fixtures; login validation, state, models and responder screens.
pytest and Requests	Actual Python agents and a live HTTP workflow against the real local API and PostgreSQL.
k6 1.3.0	Repeatable virtual-user workload with predeclared latency, failure and content thresholds.

Environment. Windows 11; .NET SDK 10.0.401 (net8.0 target); Node 22.15.0; Python 3.13.2; pytest 8.3.3; Requests 2.34.2; PostgreSQL 17.10 on 127.0.0.1:55432; API on 127.0.0.1:5009. Frontend build reported Vite 8.2.1. Flutter 3.44.9 uses its resolved lockfile; CI is pinned to that tested SDK. QMS mode requires PostgreSQL and applies EF migrations before accepting requests. Evidence records the actual tool outputs.

Isolation. A fresh ercp_qms_github database supports this full rerun. QMS startup rejects legacy tables without migration history and disables in-memory fallback. Earlier test databases are preserved. Migration tests create and remove only their own isolated databases. Most original backend tests still use EF InMemory. SMTP and Gemini credentials are disabled for live tests. No external messages or live model calls are required.

Schedule. Original analysis and testing were performed on 3 October. BUG-03 reproduction, repair, migration/backend/live workflow retesting followed by a complete reproducibility rerun and report revision were completed on 5 October. Integration with current main, publication and a full rerun were completed on 6 October. The brief states a 5 October deadline.

Consolidated Test Plan

Scope. The existing ERCP system: incident planning, resources and shelters, dispatch approval, volunteer shifts, authentication, React views, Flutter units/widgets, PostgreSQL integrity and deterministic agents. Live LLM evaluation, physical-device workflows and production-scale assurance are outside the recorded scope.

Objectives. Check normal, invalid, boundary and failure behaviour; verify a complete API workflow; measure the stated load thresholds; detect authorization and data-integrity defects; preserve source, expected results, actual results and retest evidence.

Testing Area	What is Tested	Tool/Framework	Responsible Member	Environment	Schedule
Backend/API	Controllers, services, validation and HTTP contracts	xUnit; WebApplicationFactory	All four, by component	.NET 8 test host; EF InMemory fixtures	6 Oct group; 6-7 Oct personal
Database	Constraints, relationships, transactions and migrations	xUnit; EF Core; PostgreSQL	M.Thanusikan	Isolated PostgreSQL; port 55432	5 Oct repair; 6 Oct retest
React	Components, forms, protected routes and error states	Vitest; React Testing Library	Denesh N; M.Thanusikan	Node 22; mocked API fixtures	6 Oct full rerun
Flutter	Models, forms, navigation and widget states	flutter_test	Deneshkar. P	Flutter 3.44.9; unit/widget harness	6 Oct full rerun
Integration/E2E	Incident to plan, dispatch, validation and approval	pytest; Requests; SQL verification	Jathurshy.S; Denesh N	Local API 5009 and PostgreSQL; API workflow	5 Oct repair; 6 Oct rerun
Performance	Authenticated reads; p95, errors and response checks	k6 1.3.0	Denesh N	Local API; up to 10 VUs; 40 seconds	6 Oct load run
Security	Privileged signup, role checks, JWT and hostile inputs	xUnit; pytest; Requests	Denesh N; Deneshkar. P	Test host and isolated live API	3 Oct defects; 6 Oct retest
Agentic AI	Approval, tool limits, capacity, allocation and safe failure	pytest; deterministic JSON checks	All four, matching agents	Python 3.13; live Gemini disabled	6 Oct group; 6-7 Oct personal

Names identify assigned review responsibility following the SEF component allocation. All four members review their own backend and agent cases. Shared-area ownership does not establish personal execution: dated raw records and individual appendices identify the actual operators.

Recorded testing schedule

Recorded date	Activity and retained evidence
3 October 2026	Baseline review and initial security failures: evidence/baseline and evidence/security-before.
5 October 2026	BUG-03 repair, four migration regressions and database/backend/workflow retests: evidence/migration-fix.
6 October 2026	Full group rerun: evidence/final. 270 automated passes, one k6 scenario and one migration CLI check; 272 inventory records.
6 October 2026	Deneshkar personal evidence: 4 controller, 5 HTTP and 3 safety tests; evidence/individual/IT24103828/20261006-151749.
7 October 2026	Personal runs: Jathurshy 11, Thanusikan 17 and Denesh 26 passing tests, plus two agent scenarios each; members/personal-evidence.json.
8 October 2026	Codex executed four before/after comparisons: eight scenarios passed. Deneshkar supplied a written boundary/overlap explanation; members/changed-input-evidence.json.
8 October 2026 personal runs	Four supplied terminal screenshots match the personal comparisons: two scenarios passed per member, eight total. See members/personal-changed-input-evidence.json.
Personal work remaining	All four explanations received. Members 1-2 review the documented D-series/C-series corrections. All members prepare to explain assertions and results in the viva.

Case-level expected results and assertions are in Test_Cases.html. Entry requires installed dependencies, isolated fixtures and reachable services. Exit for the evaluated scope requires recorded outcomes and retesting of demonstrated critical defects. The k6 gates are p95 below 1000 ms, HTTP failures below 1% and all content/status checks passing.

Member reruns and demonstration scenarios do not increase the 270 unique automated tests or 272 inventory records. Deneshkar supplied a written explanation on 8 October. The new comparisons were technically executed by Codex; all four personal comparison runs are supported by screenshots in Appendix I; all four explanations are received; evidence-aligned summaries and corrections are retained in Member_Explanations.md.

Test_Plan.md provides the consolidated plan and evidence paths. Changed_Input_Demonstrations.md provides runnable practice instructions and a record template; it is not completed personal evidence.

Risk based test design and responsibilities

Risk	Normal and failure or boundary coverage	Reviewer
Incorrect incident data or unsafe plans	Meaningful incident creation; invalid title/address/email and negative affected count; generated plan output and approval flag.	Jathurshy.S
Invalid resource or database state	Resource render/empty/error/status interaction; urgency levels; full shelter; exact capacity and stock limits; database unique/FK/cascade/rollback.	M.Thanusikan
Unauthorized dispatch or approval	Citizen/Volunteer mutation denial; invalid JWT; no approval before safety validation; immutable approved tasks; allocation shortages.	Denesh N
Unavailable or conflicting responders	Volunteer HTTP lifecycle; shift windows; adjacent shifts; unavailable volunteer rejection; malformed safety payload; mobile form/state tests.	Deneshkar. P

Entry criteria. Dependencies installed; isolated API and database reachable; original suites executed; valid fixtures available. **Exit criteria for the evaluated scope.** Record all outcomes, fix critical demonstrated authorization defects, verify retests, and identify remaining failures explicitly. BUG-03 now passes clean schema, rollback/data-preservation and upgrade checks. Adoption of an existing database created outside migration history remains a separate task.

Traceability. Test_Cases.html and test-cases.json contain 272 separately identified records: feature, preconditions, reproduction steps, expected assertions, actual result, status, source and evidence path. Parameterized variants are separate records. Expand the source excerpt to inspect exact fixture input and assertions.

Baseline and change. Before changes, backend 80/80 and React 39/39 passed. Seven new security regression cases produced four failures and three passes before the fix. The refreshed backend result is 95/95, including four new migration regressions. React 47/47 and Flutter 91/91 were rerun on 6 October. Existing superficial component tests remain visible and are not presented as strong behavioural coverage; the new resource and protected-route cases exercise actual components.

Ownership. The named allocation follows the user-supplied SEF ownership. Execution here was AI-assisted. Each member must review and run their contribution before claiming personal implementation or execution during the individual viva.

Integrated workflow and agent evaluation

E2E01 is a live API integration workflow. Requests drives the actual ASP.NET Core API backed by PostgreSQL. It is not a browser or physical-device end-to-end test.

Step	Observed check
1 Register citizen and report incident	201 responses; meaningful incident persisted using valid field inputs.
2 Generate response plan	Planner endpoint returned a plan for the created incident. Live LLM credentials were disabled.
3 Create responder and dispatch	Volunteer and proposed dispatch created; premature approval returned 400.
4 Assign task and validate	Task created; validation returned isValid=true and recommendation APPROVE.
5 Coordinator approves	Status update succeeded and subsequent GET returned Approved with the expected volunteer task.
6 Verify lock and persistence	Further task creation returned 409. Independent PostgreSQL query confirmed Approved.

Recorded fixture identifiers: incident 4, dispatch 1, plan 3. See evidence/final/workflow.json for the returned validation checks, saved dispatch and independent database verification. All 17 live HTTP cases passed again on 6 October against the migrated schema.

Agent evaluation. Twenty pytest cases import the actual four Python agents. Coverage includes rejected unapproved tools, coordinator structured output and mandatory human approval, allocation shortages, urgency categories, full shelters, exact capacity/stock boundaries, unavailable volunteers, malformed payload rejection and adjacent shift windows.

Injection scope. A malicious instruction string in an allocation title does not remove deterministic human approval. This only verifies that deterministic path. It is not a prompt-injection evaluation of live Gemini. Backend planner execution may use its implemented fallback; the report does not infer a live LLM response from a successful plan endpoint.

Environment recovery. The first live workflow was interrupted by Windows data-protection/event-log permissions. An authorized local rerun completed successfully. The initial failure is preserved as environment evidence and excluded from the application defect count.

Performance and security results

Performance design. The new 6 October 40-second k6 run ramps to 5 virtual users, reaches 10, then ramps down. Each iteration requests authenticated incident, dispatch and volunteer lists with a 0.2-second pause. Targets were p95 below 1000 ms, HTTP error rate below 1%, and all status/content checks passing.

Metric	Observed result
HTTP requests	3202
Throughput	79.70 requests/second
Response time p95	3.24968 ms
Maximum response time	70.90780 ms
HTTP failures	0.00%
Status and JSON checks	6402 passed; 0 failed
Thresholds	All three passed

These measurements describe a small local dataset and short authenticated read workload. They exclude mobile/browser rendering, live LLM latency, sustained stress, multi-machine deployment and production traffic. They must not be extrapolated into a production capacity claim.

Security execution. xUnit HTTP tests and pytest live HTTP tests verified public-role escalation rejection, low-privilege dispatch restrictions, malformed tokens, anonymous protected endpoints and a SQL-injection-shaped login input. The latter demonstrates rejection for the submitted input, not complete SQL-injection resistance. Coordinator success is verified through the integrated workflow.

Before and after. SEC01 initially allowed Admin and Coordinator registration (201); both now return 400. SEC02 initially let Citizen and Volunteer create dispatches (201); both now return 403. Live SEC05 additionally verifies forbidden task/status/allocation actions. Evidence is preserved in security-before.trx, backend.trx and live-api.xml.

Security limits. No ZAP scan was executed. The tested application intentionally exposes seeded demo credentials and has development-oriented configuration. These remain deployment risks. Successful authorization tests are not a complete security assessment.

Defects and quality assessment

Defect	Severity and status	Result
BUG-01 Public privileged registration	Critical / P1 Fixed and retested	DTO and public controller now reject privileged roles. Two role variants failed before and passed after.
BUG-02 Dispatch mutation permissions	High / P1 Fixed and retested	Coordinator/Admin restriction added to six mutation actions. Positive coordinator and negative low-privilege checks pass.
BUG-03 Overlapping migrations	High / P2 Fixed and retested	Duplicates and rollback ownership corrected; missing schema supplied by a new migration. Four migration regressions pass; the original failed database now upgrades.

Migration handling. The repaired history creates each table once. Rollbacks preserve earlier incident tables and resource data, and shelter foreign keys match the model. A new migration supplies Users, ValidationAudits and six missing planner fields. All eight migrations apply; the model snapshot has no pending changes. Original failures are retained alongside successful retests. Shared databases were not changed; startup-created databases require separate schema/history adoption.

Coverage. The final Coverlet report records 7567 of 9711 instrumented lines (77.92%) and 454 of 1434 branches (31.65%). Migration tests execute substantial generated/model code, so the increased line rate does not establish comparable business-logic coverage. Branch coverage remains limited. Frontend and Flutter coverage percentages were not collected; no percentages are inferred from test counts.

Static checks. Flutter analyze reports no issues. React lint reports zero errors and 4 warnings. The React production build succeeds with a bundle-size warning above 500 kB. Git diff whitespace checking passes. The team upgraded MailKit to 4.18.1 in commit cd96717. The current build has no earlier MailKit NU1902 warning; a comprehensive dependency audit is not claimed.

Remaining gaps. Physical devices, browser-driven complete workflows, live LLM failures, extended load/stress, accessibility scans, broad penetration testing and adoption of existing startup-created databases are outside the executed evidence. Existing mock-based tests must be interpreted within their isolation boundaries.

Defect_Report.html records reproduction steps, expected/actual behaviour, root cause, fix status, evidence and retest outcome for each application defect. Source changes are supplied with the application and evidence/final/security-fixes.patch and evidence/migration-fix/migration-fixes.patch.

Contribution evidence and viva preparation

Repository: github.com/Deneshkar/emergency-resource-coordination-platform

Published source: 9226a56, based on main 054fce7. The published fix/qms-submission-readiness branch preserves newer team changes. PR #47 was closed without merging; the branch remains available. Appendix A provides exact commit messages, full hashes, author/committer dates and testing contributions for every member. GitHub_Publication.md records publication and CI status.

Member	Verified existing commit and suggested demonstration
Jathurshy.S - IT24103355	50f9414: Incident controller tests and planner implementation. Demonstrate invalid input and approval requirements.
M.Thanusikan - IT24103609	25de8a9: Resource, shelter and inventory edge-case tests. Demonstrate a PostgreSQL constraint and migration regression.
Denesh N - IT24102826	48a9c3b: Dispatch allocation/controller and React tests. Explain authorization, allocation shortages and load thresholds.
Deneshkar. P - IT24103828	3c059a6: Volunteer/shift HTTP tests. Demonstrate safety rejection and a Flutter widget case.

Individual evidence status. On 7 October, Jathurshy.S, M.Thanusikan and Denesh N supplied personal-run screenshots matched to raw reports: 11, 17 and 26 tests passed, plus two agent scenarios each. Appendix G retains all six images. Deneshkar's earlier personal screenshots remain in Appendix C; his 8 October written explanation is in Appendix H. All four personal comparison runs are supported by screenshots in Appendix I; all four explanations are received; evidence-aligned summaries and corrections are retained in Member_Explanations.md.

Each member should run and explain the fixture, assertion, tool setup and result; modify an input and predict the outcome; locate one defect and its retest evidence. The marking scheme assigns 40 group marks and 60 individual marks, so reading the report alone is insufficient preparation.

AI declaration. OpenAI Codex assisted repository analysis, test ideas and implementation, security fixes, execution, evidence reconciliation and report generation on 3 October, with migration repair on 5 October and integration, publication and full retesting on 6 October 2026. Generated tests were checked against the actual source and executed. Deterministic AI paths were used with live Gemini disabled. Members must review and understand all submitted work and record their genuine contribution.

The brief requires the module CLEAR framework but does not define its fields. Use this declaration as source content for the lecturer-provided CLEAR template; no alternate definition has been assumed. Contribution_and_Viva.md provides the fuller declaration and assigned rehearsal tasks.

Conclusion and evidence index

The tested incident-to-approved-dispatch workflow completes on the actual API and PostgreSQL, and selected resource, responder, client and agent behaviours pass their automated checks. The assessment corrected two authorization defects and the migration defect. The repaired schema passes clean installation, rollback/data-preservation and upgrade checks, and the workflow passes on that schema. The new local load test met its thresholds. The remaining security, coverage and deployment-adoption limits prevent a production-readiness conclusion.

Submission item	Purpose
Software_Testing_Report.pdf	Test plan, scope, environment, results, defects and conclusion.
Test_Cases.html and test-cases.json	272 completed, evidence-linked records with expected and actual results.
Defect_Report.html and defects.json	Three defects with severity, reproduction, evidence and retest status.
Contribution_Record.md, Contribution_and_Viva.md and GitHub_Publication.md	Verified commit links, publication/CI status, personal execution evidence, viva guide and AI declaration.
Reproduction_Guide.md and scripts/run-tests.ps1	Dependencies, isolated environment setup and rerun commands.
evidence/baseline and evidence/security-before	Original suites and pre-fix security failures.
evidence/final/backend, react.json, flutter.jsonl, agents.xml, live-api.xml	Raw automated execution and backend coverage.
evidence/final/performance.json, workflow.json and migrations.log	Load metrics, migrated workflow evidence and passing CLI upgrade. Original failure and four regression retests are in evidence/migration-fix.
Application source and testing scripts	Reproducible implementation, configuration examples and test code.

Submission checks. Review member details; supply a group number if required; complete genuine individual review/run evidence; transfer the AI declaration into the official CLEAR format; rehearse the individual demonstrations; upload through the correct CourseWeb entry. The QMS source is published on fix/qms-submission-readiness; PR #47 is closed without merging. CourseWeb upload has not been performed.

Source basis. The supplied four-page SE3110 Assignment.pdf, the ERCP repository at published source commit 9226a56 on top of main 054fce7, and the tool-generated evidence listed above. Historical report pass counts were not reused as fresh execution results.

Appendix A.1 - Individual contribution

Jathurshy.S | IT24103355

SEF responsibility: Incident Management and Coordinator/Planner Agent

Verified GitHub account: Jathurshy11

The following are historical contributions in the tested project. Dates are recorded in Sri Lanka time (UTC+05:30). Author and committer dates are shown separately because rebasing or integration can change the latter.

Commit: 50f9414e8ceeb8f287112bde2a81883fa9ba4245

Exact message: feat: complete student 1 incident intelligence and planner agent

Author date: 2026-09-16T10:55:54+05:30
Committer date: 2026-09-21T12:29:48+05:30

Testing / repair contribution: Incident controller tests accompany incident input handling and coordinator/planner implementation. This is backend test authorship evidence.

Relevant committed files:
backend-tests/IncidentControllerTests.cs

Individual summary. The Git history above supports implementation/test authorship. A submitted personal run on 7 October is supported by terminal screenshots and matching raw reports in Appendix G. Personal changed-input screenshots and raw reports are now supplied in Appendix I. Written explanations are received; the reconciled summaries are retained in Member_Explanations.md. Wider database, live workflow and performance review is not inferred from this focused run.

Screenshot evidence: See Appendix B.1 for the user-supplied actual GitHub commit page. The unchanged original PNG is included in evidence/github.

Appendix A.2 - Individual contribution

M.Thanusikan | IT24103609

SEF responsibility: Resource and Inventory Management and Situation Analysis Agent

Verified GitHub account: it24103609

The following are historical contributions in the tested project. Dates are recorded in Sri Lanka time (UTC+05:30). Author and committer dates are shown separately because rebasing or integration can change the latter.

Commit: 25de8a92bee86fe982b6fdbbe65afdc35055ce43

Exact message: test(backend): Add edge case unit tests for shelters, inventory, and geolocation

Author date: 2026-09-16T08:24:46+05:30
Committer date: 2026-09-16T08:24:46+05:30

Testing / repair contribution: Backend edge-case tests cover shelters, inventory and geolocation. These committed tests are distinct from the later QMS PostgreSQL and migration regressions.

Relevant committed files:
backend-tests/GeolocationServiceTests.cs
backend-tests/InventoryControllerTests.cs
backend-tests/SheltersControllerTests.cs

Commit: 6846af6080a41d8f9f02be333440d2fdcb641714

Exact message: test(react): Add unit tests for ShelterList, ResourceList, and InventoryList pages

Author date: 2026-09-16T08:28:53+05:30
Committer date: 2026-09-16T08:28:53+05:30

Testing / repair contribution: React page tests cover ShelterList, ResourceList and InventoryList. Existing component tests retain the scope and limitations described in the report.

Relevant committed files:
frontend-react/src/pages/__tests__/InventoryList.test.jsx
frontend-react/src/pages/__tests__/ResourceList.test.jsx
frontend-react/src/pages/__tests__/ShelterList.test.jsx

Individual summary. The Git history above supports implementation/test authorship. A submitted personal run on 7 October is supported by terminal screenshots and matching raw reports in Appendix G. Personal changed-input screenshots and raw reports are now supplied in Appendix I. Written explanations are received; the reconciled summaries are retained in Member_Explanations.md. Wider database, live workflow and performance review is not inferred from this focused run.

Screenshot evidence: See Appendix B.2 for the user-supplied actual GitHub commit page. The unchanged original PNG is included in evidence/github.

Appendix A.3 - Individual contribution

Denesh N | IT24102826

SEF responsibility: Dispatch and Task Assignment and Resource Allocation Agent

Verified GitHub account: Denesh005

The following are historical contributions in the tested project. Dates are recorded in Sri Lanka time (UTC+05:30). Author and committer dates are shown separately because rebasing or integration can change the latter.

Commit: 48a9c3b9cd87d67ce48b9c16aeb960be85baa266

Exact message: Implement dispatch task allocation

Resolved resource ID consistency, secured the dispatch approval workflow, completed the coordinator dispatch UI flow and added allocation and validation tests.

Author date: 2026-09-26T12:35:36+05:30
Committer date: 2026-09-26T12:35:36+05:30

Testing / repair contribution: Backend dispatch allocation, validation and controller tests, plus React dispatch-page and validation-service tests, accompany the dispatch implementation.

Relevant committed files:
backend-tests/DispatchAllocationTests.cs
backend-tests/DispatchValidationServiceTests.cs
backend-tests/DispatchesControllerTests.cs
frontend-react/src/pages/__tests__/DispatchList.test.jsx
frontend-react/src/services/__tests__/validationService.test.js

Commit: cd967173f349ae343a729b885cb7182eca7a2d83

Exact message: fix: database migrations and prepare backend deployment

Migration: EF Core database updates for safe schema changes.

Author date: 2026-10-05T12:30:00+05:30
Committer date: 2026-10-05T12:30:00+05:30

Testing / repair contribution: Removed duplicate migration table operations and upgraded MailKit. This is a database repair contribution; this commit does not itself prove execution of the later QMS migration tests.

Relevant committed files:
backend/ERCP.Api.csproj
backend/Migrations/20260916110602_AddAffectedCountToIncidents.cs

Individual summary. The Git history above supports implementation/test authorship. A submitted personal run on 7 October is supported by terminal screenshots and matching raw reports in Appendix G. Personal changed-input screenshots and raw reports are now supplied in Appendix I. Written explanations are received; the reconciled summaries are retained in Member_Explanations.md. Wider database, live workflow and performance review is not inferred from this focused run.

Screenshot evidence: See Appendix B.3 for the user-supplied actual GitHub commit page. The unchanged original PNG is included in evidence/github.

Appendix A.4 - Individual contribution

Deneshkar. P | IT24103828

SEF responsibility: Volunteer/Responder Management and Validation/Safety Agent

Verified GitHub account: Deneshkar

The following are historical contributions in the tested project. Dates are recorded in Sri Lanka time (UTC+05:30). Author and committer dates are shown separately because rebasing or integration can change the latter.

Commit: 3c059a63c9093f6347de4fbef73f9b3e7164910a

Exact message: feat: add volunteer shift and API integration tests, and update documentation

Author date: 2026-09-16T21:08:35+05:30
Committer date: 2026-09-21T12:29:49+05:30

Testing / repair contribution: Volunteer shift tests and volunteer API integration tests accompany responder validation and controller changes.

Relevant committed files:
backend-tests/VolunteerShiftTests.cs
backend-tests/VolunteersApiIntegrationTests.cs

Commit: ee924e23010cdc591afd7fa393c3b239c438bfef

Exact message: feat(security): restrict public registration to Citizen/Volunteer, lock dispatch writes to Coordinator/Admin

Author date: 2026-10-03T13:21:27+05:30
Committer date: 2026-10-03T13:21:27+05:30

Testing / repair contribution: Committed public-role and dispatch authorization fixes, PostgreSQL/security regressions, and React protected-route/resource behaviour tests. QMS preparation and execution were AI-assisted.

Relevant committed files:
backend-tests/AuthControllerTests.cs
backend-tests/QmsPostgresTests.cs
backend-tests/QmsSecurityTests.cs
backend-tests/VolunteersApiIntegrationTests.cs
frontend-react/src/components/__tests__/ProtectedRoute.test.jsx
frontend-react/src/pages/__tests__/ResourceBehaviour.test.jsx

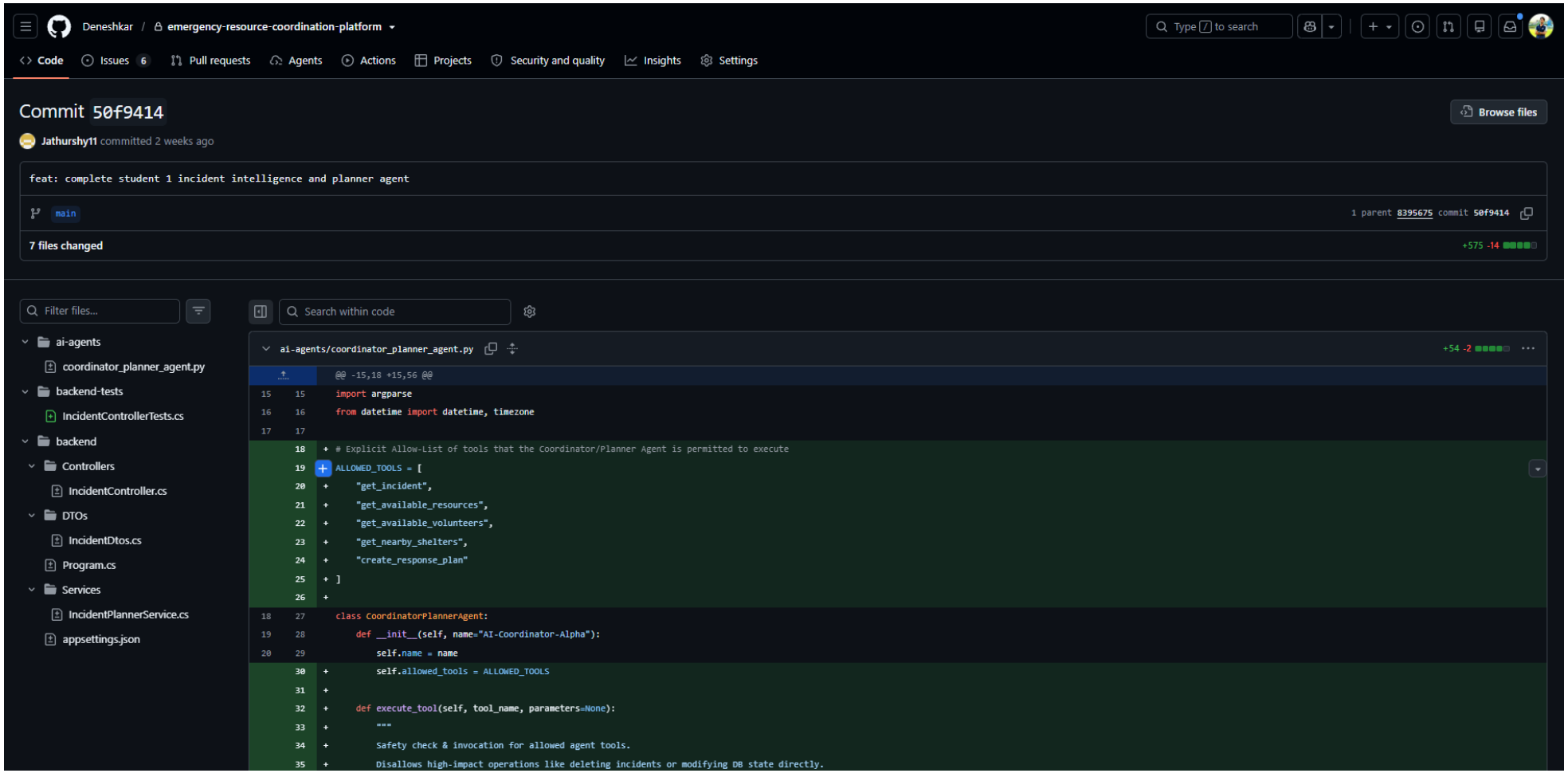
Individual summary. Contributed volunteer/shift API tests and published security/QMS changes. Group analysis, execution, integration and reporting used OpenAI Codex assistance under the repository owner's direction. Personal viva review and demonstrations must be recorded separately.

Personal execution confirmed: Four controller, five HTTP integration and three safety-agent tests passed during guided practice on 6 October. Six supplied screenshots and raw reports are in Appendix C. These are existing test reruns; own-word explanations and the boundary exercise remain unconfirmed.

Screenshot evidence: See Appendix B.4 for the user-supplied actual GitHub commit page. The unchanged original PNG is included in evidence/github.

Appendix B.1 - GitHub commit screenshot

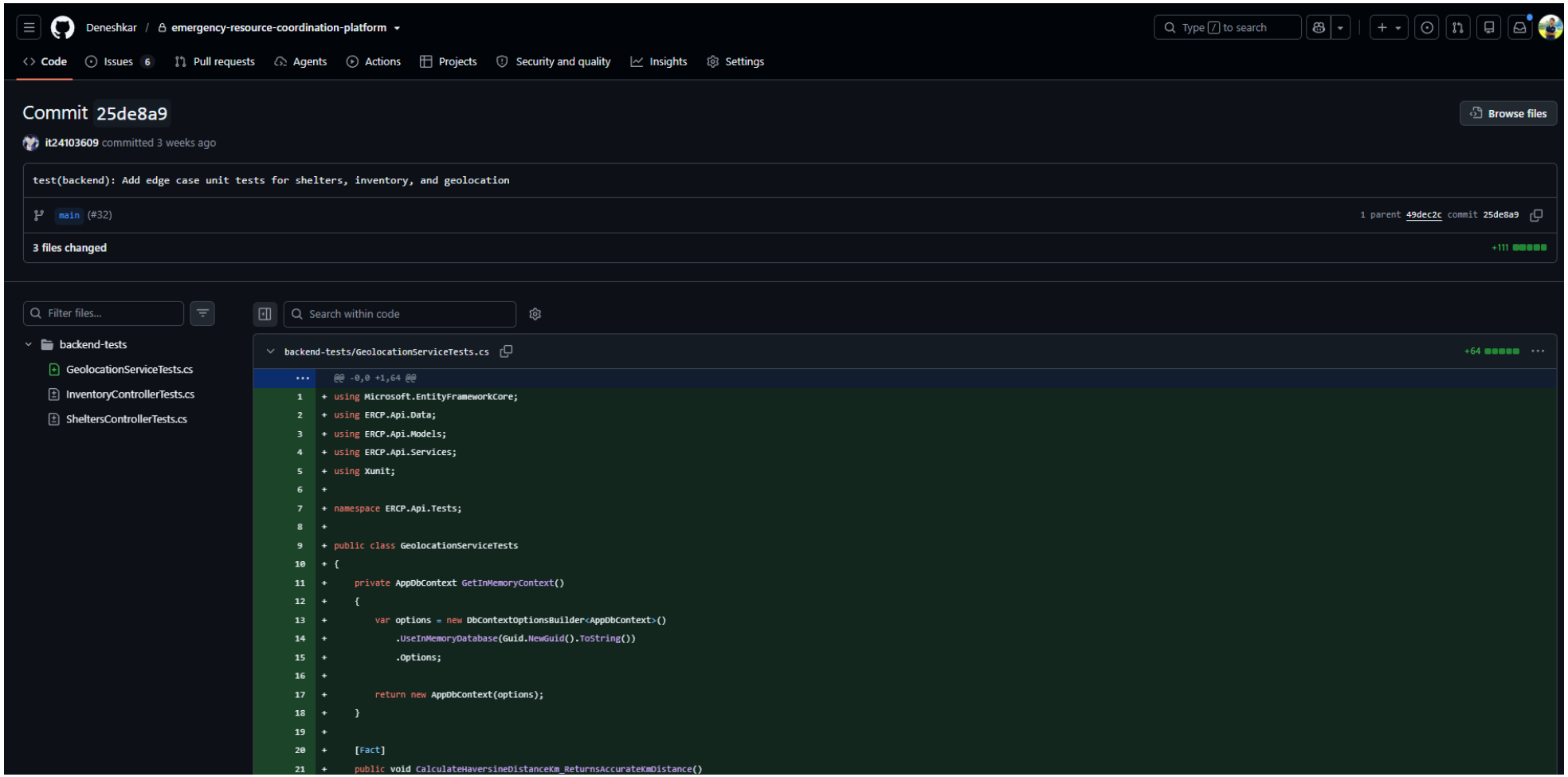
Jathurshy.S | IT24103355 - 50f9414e8ceeb8f287112bde2a81883fa9ba4245



Actual GitHub page screenshot supplied by the group on 2026-10-06. Original PNG preserved unchanged; shown proportionally scaled. Full commit messages and exact dates are in Appendix A. Original: evidence/github/Jathurshy-50f9414.png.

Appendix B.2 - GitHub commit screenshot

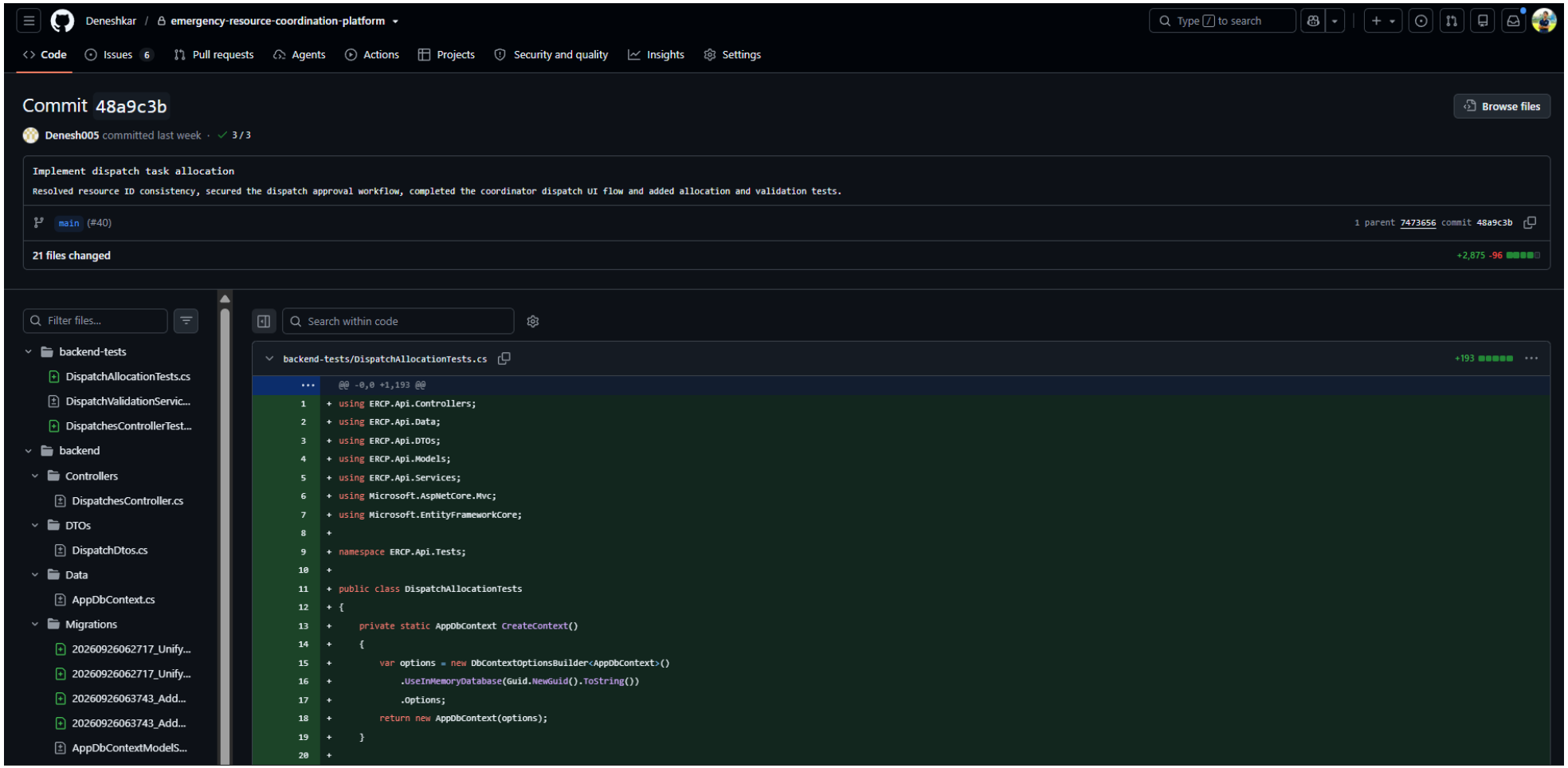
M.Thanusikan | IT24103609 - 25de8a92bee86fe982b6fdbbe65afdc35055ce43



Actual GitHub page screenshot supplied by the group on 2026-10-06. Original PNG preserved unchanged; shown proportionally scaled. Full commit messages and exact dates are in Appendix A. Original: evidence/github/Thanusikan-25de8a9.png.

Appendix B.3 - GitHub commit screenshot

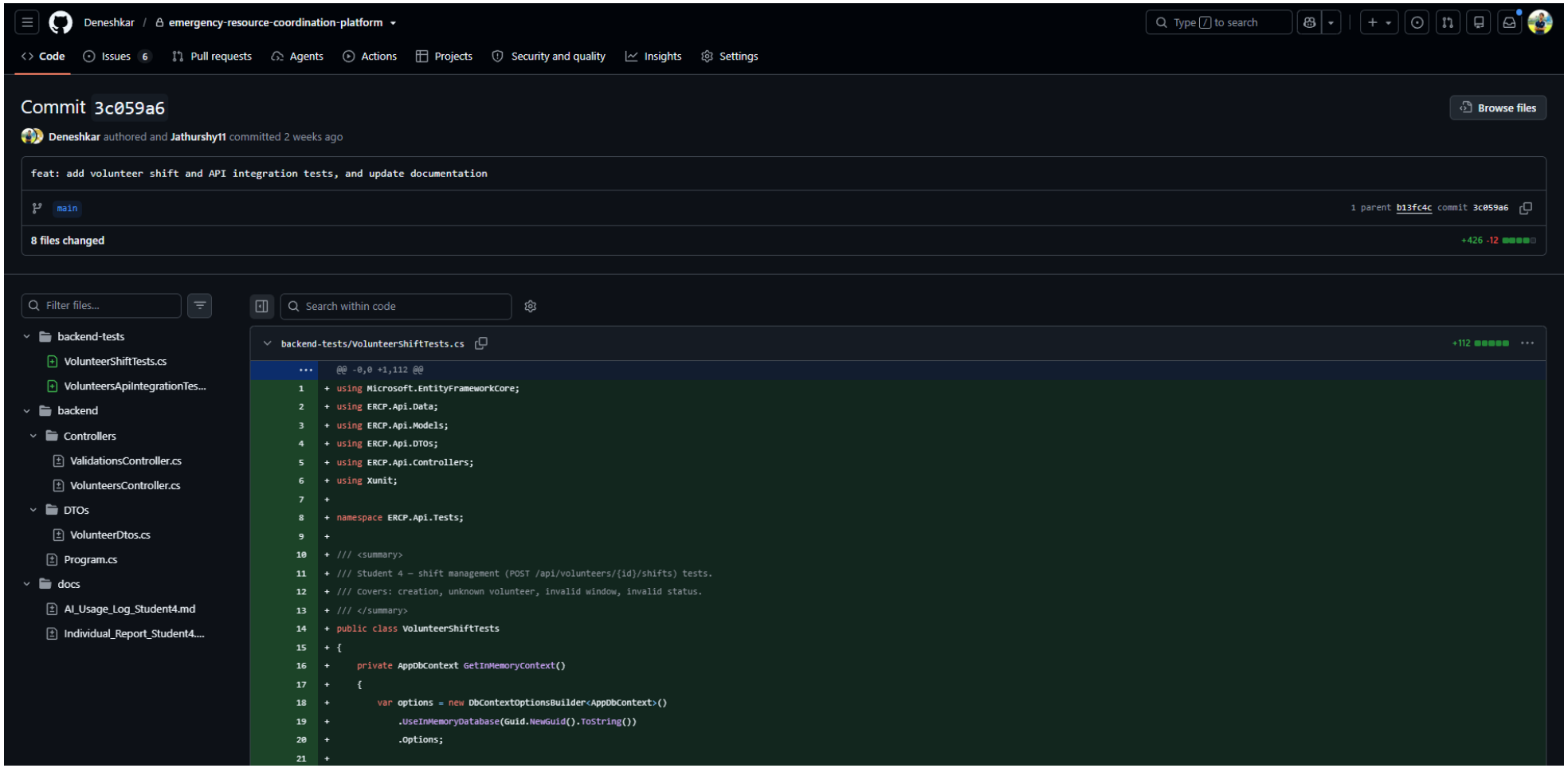
Denesh N | IT24102826 - 48a9c3b9cd87d67ce48b9c16aeb960be85baa266



Actual GitHub page screenshot supplied by the group on 2026-10-06. Original PNG preserved unchanged; shown proportionally scaled. Full commit messages and exact dates are in Appendix A. Original: evidence/github/Denesh-48a9c3b.png.

Appendix B.4 - GitHub commit screenshot

Deneshkar. P | IT24103828 - 3c059a63c9093f6347de4fbef73f9b3e7164910a



Actual GitHub page screenshot supplied by the group on 2026-10-06. Original PNG preserved unchanged; shown proportionally scaled. Full commit messages and exact dates are in Appendix A. Original: evidence/github/Deneshkar-3c059a6.png.

GitHub identifies Deneshkar as author and Jathurshy11 as committer. These are distinct Git roles; the volunteer contribution is attributed to the recorded author.

Appendix C - Personal execution evidence

Deneshkar. P | IT24103828 | 6 October 2026

The student personally ran the commands during guided practice and supplied six terminal screenshots. OpenAI Codex assisted with command selection, explanations and evidence organization. Original screenshots are preserved unchanged, and saved TRX/XML reports independently confirm the results.

Personally executed area	Passed	Failed	Skipped
Volunteer controller tests	4	0	0
Volunteer HTTP integration tests	5	0	0
Deterministic safety-agent tests	3	0	0

These 12 executions rerun existing cases. Group totals remain 270 automated cases and 272 inventory records. The 17 agent cases outside the selected filter were intentionally deselected, not skipped or failed.

The controller and internal HTTP test-server fixtures use EF Core InMemory. They do not establish PostgreSQL constraints. The Python safety tests exercise deterministic rules without a live LLM.

Volunteer controller tests
Started: 2026-10-06T15:18:01.0555416+05:30
Raw report: evidence/individual/IT24103828/20261006-151749/volunteer-shifts.trx

Volunteer HTTP integration tests
Started: 2026-10-06T15:21:23.3335801+05:30
Raw report: evidence/individual/IT24103828/20261006-151749/volunteer-http.trx

Deterministic safety-agent tests
Started: 2026-10-06T15:23:50.152024+05:30
Raw report: evidence/individual/IT24103828/20261006-151749/safety-agent-20261006-152348.xml

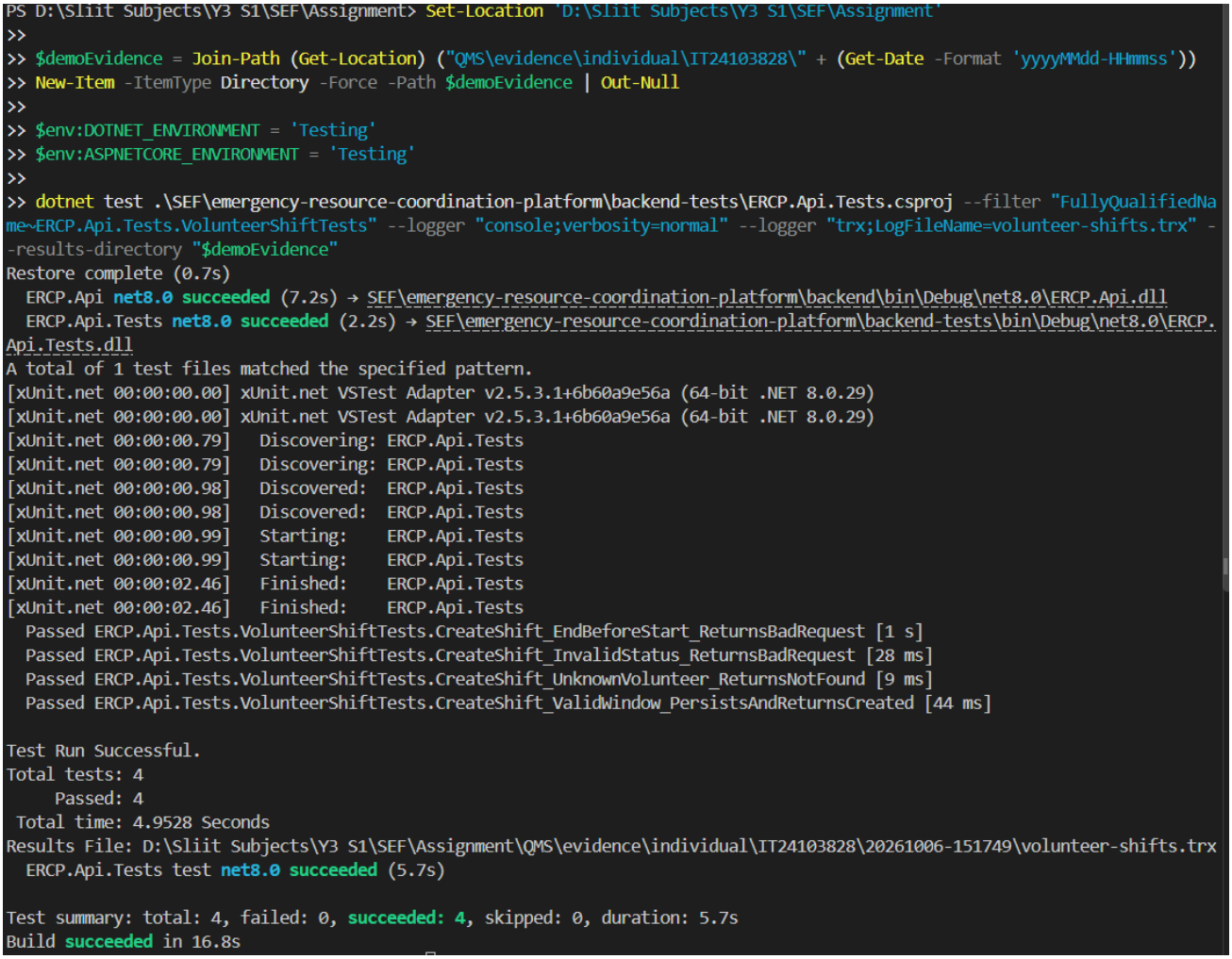
The HTTP results screenshot reports overwriting volunteer-http.trx. The current report is the retained latest execution; the four HTTP screenshots are supporting views, not four additional test suites.

8 October update: Deneshkar supplied a written boundary/overlap explanation in Appendix H and a personal changed-input terminal screenshot with matching reports in Appendix I. His new comparison passed both scenarios. Defect/retest interpretation and the assessed viva remain separate preparation tasks. The other three members' submitted 7 October personal runs are recorded in Appendix G.

Individual_Evidence_IT24103828.md links all six PNGs, the raw results and personal-execution.json, which records their SHA-256 hashes. The following six pages reproduce every supplied personal test screenshot.

Appendix C.1 - Controller test results

Deneshkar. P | IT24103828 - Guided personal testing, 6 October 2026

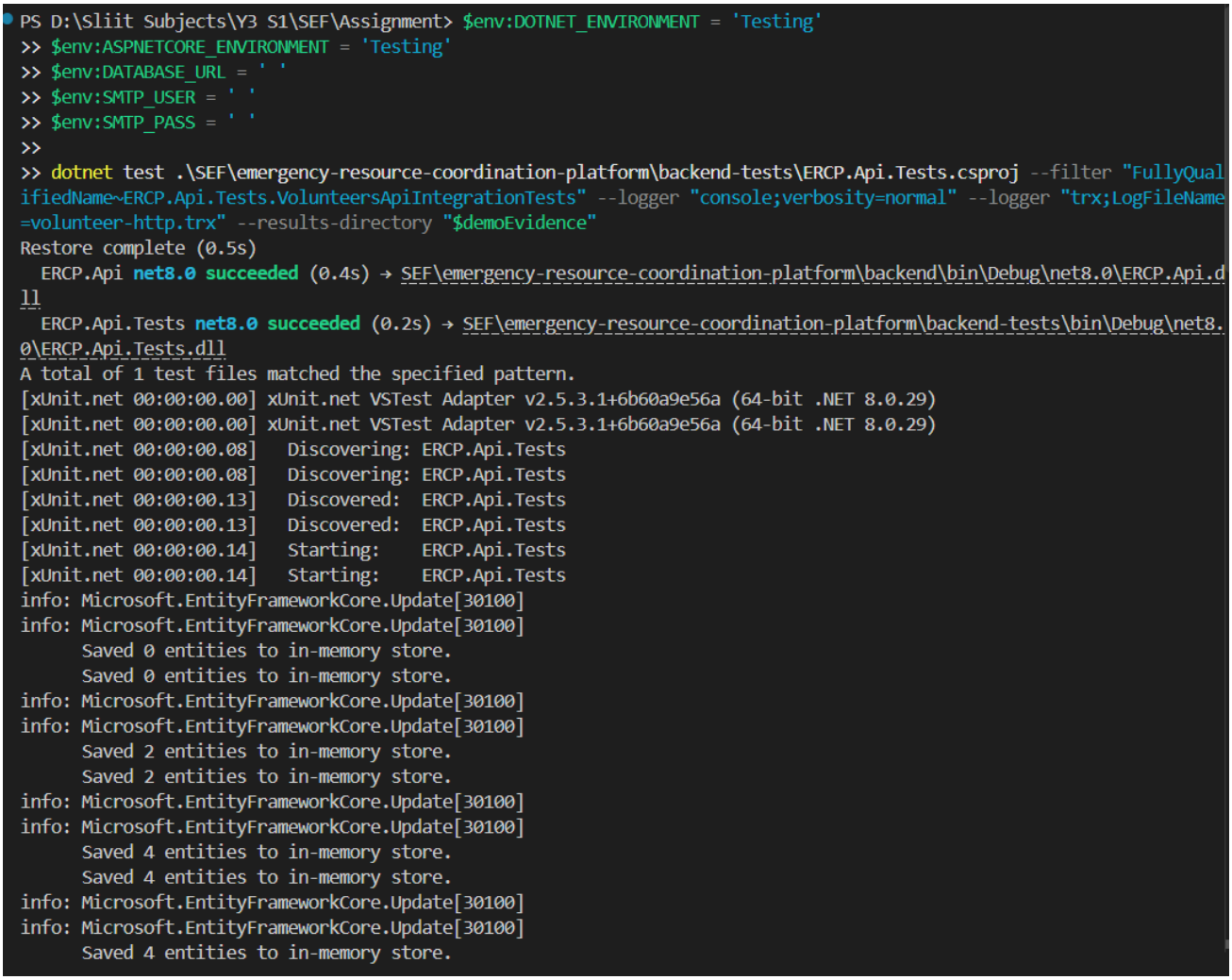


User-supplied screenshot, preserved unchanged and scaled proportionally. Supporting raw report: volunteer-shifts.trx.

Original PNG: evidence/individual/IT24103828/20261006-151749/01-volunteer-shifts-results.png

Appendix C.2 - HTTP test command and startup

Deneshkar. P | IT24103828 - Guided personal testing, 6 October 2026



User-supplied screenshot, preserved unchanged and scaled proportionally. Supporting raw report: volunteer-http.trx.

Original PNG: evidence/individual/IT24103828/20261006-151749/02-volunteer-http-command-and-start.png

Appendix C.3 - HTTP test in-memory database log

Deneshkar. P | IT24103828 - Guided personal testing, 6 October 2026

```
PS D:\Sliit Subjects\Y3 S1\SEF\Assignment> $env:DOTNET_ENVIRONMENT = 'Testing'
>> $env:ASPNETCORE_ENVIRONMENT = 'Testing'
>> $env:DATABASE_URL = ' '
>> $env:SMTP_USER = ' '
>> $env:SMTP_PASS = ' ' …
info: Microsoft.EntityFrameworkCore.Update[30100]
      Saved 4 entities to in-memory store.
      Saved 4 entities to in-memory store.
info: Microsoft.EntityFrameworkCore.Update[30100]
info: Microsoft.EntityFrameworkCore.Update[30100]
      Saved 2 entities to in-memory store.
      Saved 2 entities to in-memory store.
info: Microsoft.EntityFrameworkCore.Update[30100]
info: Microsoft.EntityFrameworkCore.Update[30100]
      Saved 1 entities to in-memory store.
      Saved 1 entities to in-memory store.
info: Microsoft.EntityFrameworkCore.Update[30100]
info: Microsoft.EntityFrameworkCore.Update[30100]
      Saved 5 entities to in-memory store.
      Saved 5 entities to in-memory store.
info: Microsoft.EntityFrameworkCore.Update[30100]
info: Microsoft.EntityFrameworkCore.Update[30100]
      Saved 8 entities to in-memory store.
      Saved 8 entities to in-memory store.
info: Microsoft.EntityFrameworkCore.Update[30100]
info: Microsoft.EntityFrameworkCore.Update[30100]
      Saved 8 entities to in-memory store.
      Saved 8 entities to in-memory store.
info: Microsoft.EntityFrameworkCore.Update[30100]
info: Microsoft.EntityFrameworkCore.Update[30100]
      Saved 1 entities to in-memory store.
      Saved 1 entities to in-memory store.
info: Microsoft.EntityFrameworkCore.Update[30100]
info: Microsoft.EntityFrameworkCore.Update[30100]
      Saved 1 entities to in-memory store.
      Saved 1 entities to in-memory store.
info: ERCP.Api.Services.AuthService[0]
info: ERCP.Api.Services.AuthService[0]
```

User-supplied screenshot, preserved unchanged and scaled proportionally. Supporting raw report: volunteer-http.trx.

Original PNG: evidence/individual/IT24103828/20261006-151749/03-volunteer-http-database-log.png

Appendix C.4 - HTTP test authentication setup log

Deneshkar. P | IT24103828 - Guided personal testing, 6 October 2026

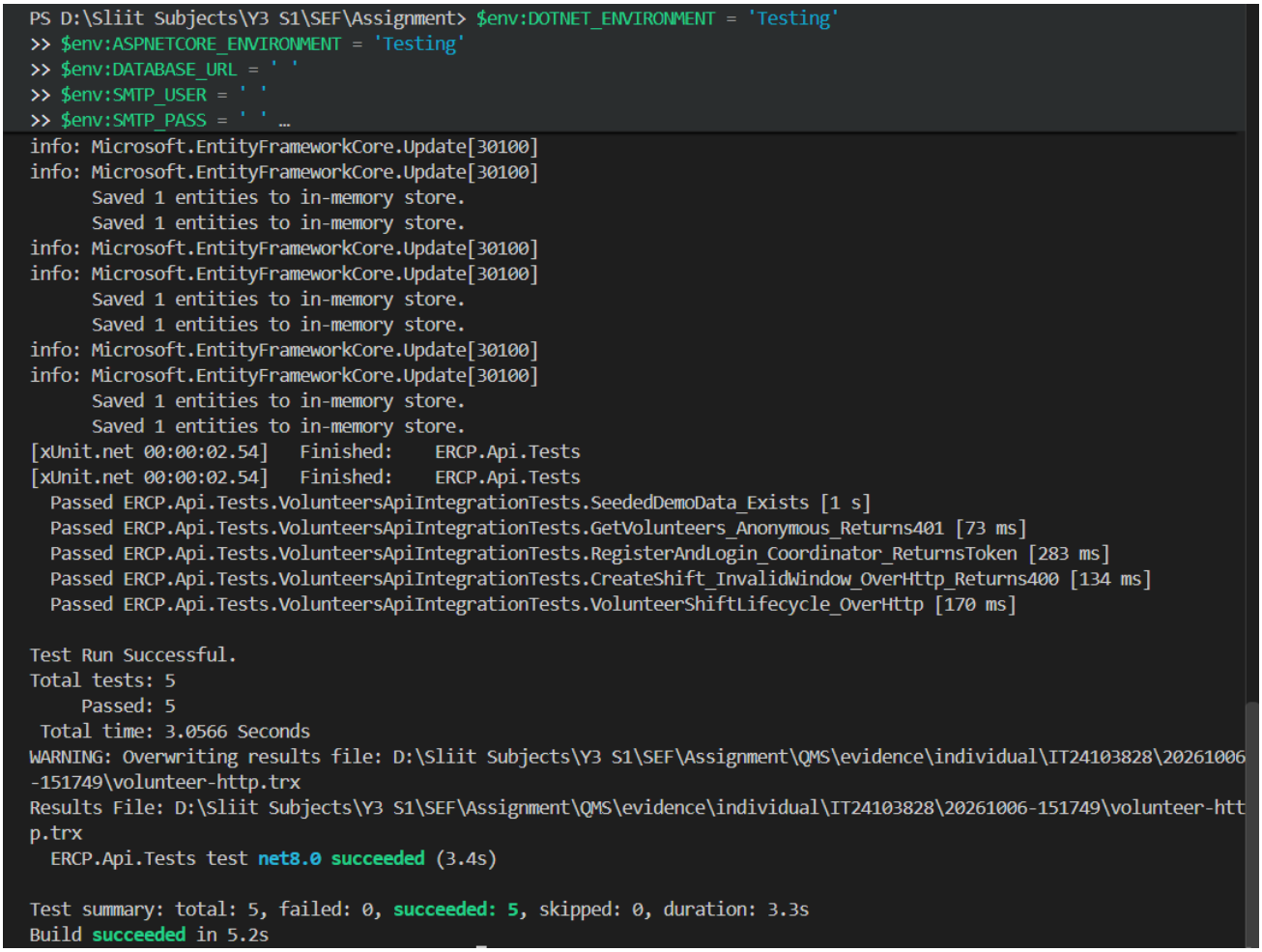
```
PS D:\Sliit Subjects\Y3 S1\SEF\Assignment> $env:DOTNET_ENVIRONMENT = 'Testing'
>> $env:ASPNETCORE_ENVIRONMENT = 'Testing'
>> $env:DATABASE_URL = ' '
>> $env:SMTP_USER = ' '
>> $env:SMTP_PASS = ' ' …
      New user registered: e2e-coord-3de343a0-fa7f-46f0-ab19-02a61fada22b@test.local with role Citizen
      New user registered: e2e-coord-3de343a0-fa7f-46f0-ab19-02a61fada22b@test.local with role Citizen
info: Microsoft.EntityFrameworkCore.Update[30100]
info: Microsoft.EntityFrameworkCore.Update[30100]
      Saved 1 entities to in-memory store.
      Saved 1 entities to in-memory store.
info: Microsoft.EntityFrameworkCore.Update[30100]
info: Microsoft.EntityFrameworkCore.Update[30100]
      Saved 1 entities to in-memory store.
      Saved 1 entities to in-memory store.
info: ERCP.Api.Services.AuthService[0]
info: ERCP.Api.Services.AuthService[0]
      New user registered: e2e-coord-564e557d-a511-4a46-b1b7-648970473eff@test.local with role Citizen
      New user registered: e2e-coord-564e557d-a511-4a46-b1b7-648970473eff@test.local with role Citizen
info: Microsoft.EntityFrameworkCore.Update[30100]
info: Microsoft.EntityFrameworkCore.Update[30100]
      Saved 1 entities to in-memory store.
      Saved 1 entities to in-memory store.
info: Microsoft.EntityFrameworkCore.Update[30100]
info: Microsoft.EntityFrameworkCore.Update[30100]
      Saved 1 entities to in-memory store.
      Saved 1 entities to in-memory store.
info: Microsoft.EntityFrameworkCore.Update[30100]
info: Microsoft.EntityFrameworkCore.Update[30100]
      Saved 1 entities to in-memory store.
      Saved 1 entities to in-memory store.
info: ERCP.Api.Services.AuthService[0]
info: ERCP.Api.Services.AuthService[0]
      New user registered: e2e-coord-e9a8e92f-89ad-4a01-a838-63a05adf2041@test.local with role Citizen
      New user registered: e2e-coord-e9a8e92f-89ad-4a01-a838-63a05adf2041@test.local with role Citizen
info: Microsoft.EntityFrameworkCore.Update[30100]
info: Microsoft.EntityFrameworkCore.Update[30100]
      Saved 1 entities to in-memory store.
```

User-supplied screenshot, preserved unchanged and scaled proportionally. Supporting raw report: volunteer-http.trx.

Original PNG: evidence/individual/IT24103828/20261006-151749/04-volunteer-http-authentication-log.png

Appendix C.5 - HTTP test results

Deneshkar. P | IT24103828 - Guided personal testing, 6 October 2026

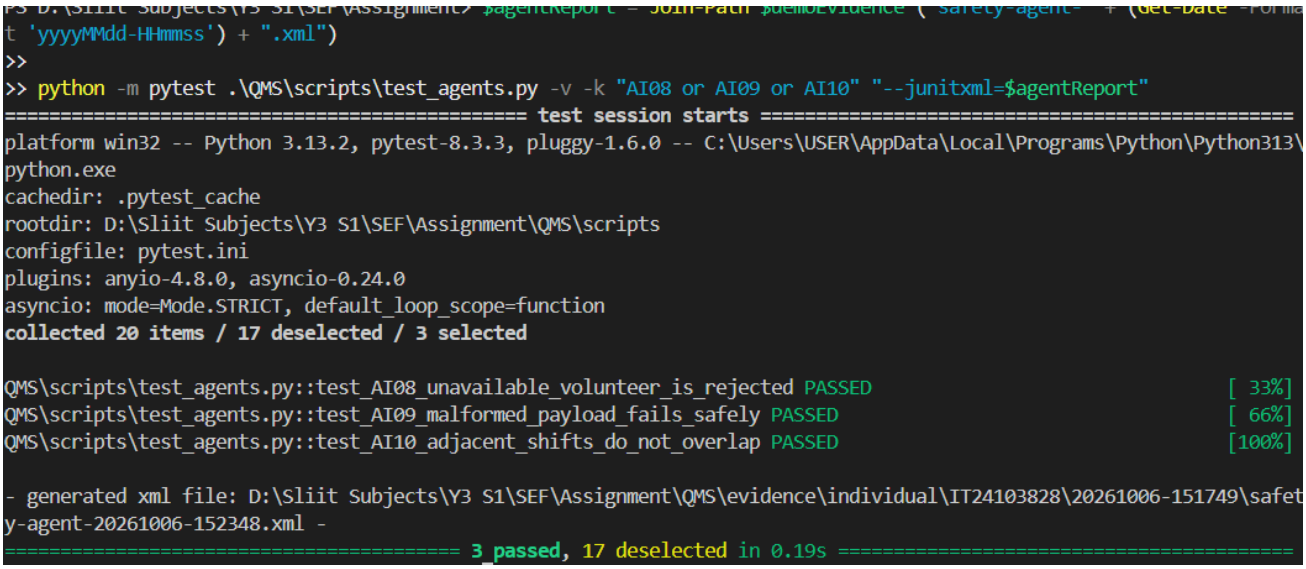


User-supplied screenshot, preserved unchanged and scaled proportionally. Supporting raw report: volunteer-http.trx.

Original PNG: evidence/individual/IT24103828/20261006-151749/05-volunteer-http-results.png

Appendix C.6 - Safety-agent test results

Deneshkar. P | IT24103828 - Guided personal testing, 6 October 2026



User-supplied screenshot, preserved unchanged and scaled proportionally. Supporting raw report: safety-agent-20261006-152348.xml.

Original PNG: evidence/individual/IT24103828/20261006-151749/06-safety-agent-results.png

Appendix D - Individual demo packages

Open members/index.html in a local browser to choose a member. Each package contains a test-case document with original source links, a named PowerShell test runner, editable agent inputs, an actual agent output report and dated execution evidence. The complete source repository or assignment ZIP is required.

Member / ID	Business component and matching agent	Tests	Demos
Jathurshy.S IT24103355	Incident Management and Coordinator/Planner Agent	9 BE 2 AI	2
M.Thanusikan IT24103609	Resource and Inventory Management and Situation Analysis Agent	12 BE 5 AI	2
Denesh N IT24102826	Dispatch and Task Assignment and Resource Allocation Agent	23 BE 3 AI	2
Deneshkar. P IT24103828	Volunteer/Responder Management and Validation/Safety Agent	9 BE 10 AI	4

Package verification. Codex executed all four packages on 6 October: 73 selected existing tests (53 backend and 20 agent cases) and 10 separately checked demonstration scenarios passed. Runs are labelled ai-assisted and linked by members/verification.json. An intentionally wrong safety expectation also returned failure while retaining the actual output.

Counts and scope. These are reruns of selected group cases, not additional unique tests. The group totals remain 270 automated cases and 272 inventory records. Selected backend fixtures use InMemory/TestServer, while agent demonstrations call the actual Python classes in deterministic mode. The wider PostgreSQL, frontend, mobile, live API and performance evidence remains in the group suite.

Present each member's work. Open that member's index.html, show Test_Cases.html, run run-memberN-tests.ps1 with -ExecutionKind personal and -Operator followed by the actual name, and open the new results.html. Then show agent-inputs.json and agent-demo.html, explain the returned decision/checks, change an input and predict the next output.

Business-specific agent outputs. Member 1 demonstrates incident plans and mandatory human approval. Member 2 demonstrates urgency, inventory deficits and shelter capacity. Member 3 demonstrates covered/shortage allocation proposals. Member 4 demonstrates available/unavailable responders and overlapping/adjacent shifts. Recommendations do not themselves approve or execute dispatches. Situation analysis selects by available shelter capacity, not geographical distance.

Personal evidence remains separate. Deneshkar's earlier 12-test personal practice and six screenshots remain in Appendix C. The full member-4 package selects 19 existing tests; its technical verification does not establish a personal 19-test run. Members 1 to 3 submitted personal execution screenshots on 7 October. Deneshkar supplied a written explanation on 8 October; all four personal comparison runs are supported by screenshots in Appendix I; all four explanations are received; evidence-aligned summaries and corrections are retained in Member_Explanations.md.

Member_Demonstrations.md provides the four package links and commands. The package scripts and presentation pages were prepared with OpenAI Codex assistance; member ownership labels do not reassign original Git authorship.

Appendix E - Member 4 assisted execution

Deneshkar. P | IT24103828
Volunteer/Responder Management and Validation/Safety Agent

Actual operator: OpenAI Codex, on behalf of Deneshkar. P, at the student's request.
Started: 06 October 2026, 22:39:20 (Asia/Colombo, UTC+05:30). This is an AI-assisted technical run, not a newly confirmed personal execution or assessment of understanding.

Executed command from the member folder:
.\run-member4-tests.ps1 -ExecutionKind ai-assisted
-Operator "OpenAI Codex (on behalf of Deneshkar. P)"

Executed area	Passed	Failed	Skipped
Volunteer shift controller	4	0	0
Volunteer HTTP integration	5	0	0
Deterministic safety-agent tests	10	0	0
Separate agent demonstration scenarios	4	0	0

Actual agent input and output. The runner calls ValidationSafetyAgent.validate_assignment. All shift times below are UTC on 6 October 2026; scenario inputs and complete outputs are preserved as JSON.

Scenario input	Observed output	Check
M4-D01 Available: true; proposed 10:00-11:00; existing none	is_valid=true APPROVE	PASS
M4-D02 Available: false; proposed 10:00-11:00; existing none	is_valid=false REJECT	PASS
M4-D03 Available: true; proposed 09:30-11:00; existing 09:00-10:00	is_valid=false REVISE	PASS
M4-D04 Available: true; proposed 10:00-11:00; existing 09:00-10:00	is_valid=true APPROVE	PASS

Evidence folder: members/member4-IT24103828/results/ai-assisted-20261006-223920-978734
results.html, backend.trx, agents.xml, execution.json, agent-inputs.json, agent-output.json and agent-demo.html.

Scope. Nineteen selected existing tests passed, with four demo scenarios checked separately. Group totals remain 270 automated cases and 272 inventory records. Backend fixtures use InMemory/TestServer; the Python agent uses deterministic rules, with no live LLM. APPROVE is a recommendation, not a saved dispatch approval.

Personal evidence and understanding. Appendix C retains the student's earlier 12 personally executed tests and six supplied screenshots. The 19-test run above was executed by Codex. The student supplied a written boundary/overlap explanation on 8 October, retained in Appendix H. His personal changed-input run is supported by a supplied screenshot and matching reports in Appendix I.

Separate declared-personal record. The results folder personal-20261006-223824-403789 records Deneshkar. P as operator at 22:38 Asia/Colombo. Its raw reports confirm 19 passing tests and four passing demos. This run was found separately from the Codex run above; confirmation of who operated it is pending. The personal label alone does not establish understanding.

Appendix F.1 - Member 1 technical execution

Jathurshy.S | IT24103355
Incident Management and CoordinatorPlannerAgent

Historical technical execution - run by Codex; later personal evidence and explanations are recorded separately.
Operator: OpenAI Codex. Started 06 October 2026, 22:56:42 (Asia/Colombo, UTC+05:30). This historical record documents component tests and actual agent behaviour. Later submitted personal runs appear in Appendix G; written explanations are now recorded in Member_Explanations.md; viva assessment remains separate.

Runner: run-member1-tests.ps1. Exact commands, operator and timestamps are retained in execution.json.

Test area	Passed	Failed	Skipped
IncidentControllerTests	7	0	0
IncidentPlannerServiceTests	2	0	0
Python agent tests	2	0	0
Total selected existing tests	11	0	0
Separate agent demonstration scenarios	2	0	0

M1-D01 - Incident with available responders

Input: Incident 7; priority High; resources 1; volunteers 1.

Observed: incidentId = 7; requiresHumanApproval = true; actions = ["Secure perimeter at Demo Sector", "Deploy allocated resources and volunteers"]
Scenario result: PASS

M1-D02 - Incident without available resources

Input: Incident 8; priority Medium; resources 0; volunteers 0.

Observed: requiresHumanApproval = true; reasoning = 1. Observed Medium priority General incident with 2 affected.
2. Gathered real data: No available resources found.
3. Gathered real data: No available volunteers found.
4. Identified constraint: Lack of required vehicles.
5. Identified constraint: Lack of required volunteers.
6. Formulating safe dispatch plan. Requires human validation before execution.
Scenario result: PASS

Evidence folder: members/member1-IT24103355/results/ai-assisted-20261006-225642-874010
Open results.html for named test outcomes and agent-demo.html for full inputs and actual outputs. Raw backend.trx, agents.xml, logs, source hashes and JSON records are included.

Scope. These are reruns of existing cases, not additional unique tests; group totals remain 270 automated cases and 272 inventory records. Selected backend fixtures use InMemory/TestServer and agent demos use the real Python classes in deterministic mode, without live LLM calls.

Incident details and supplied resources produce a response plan with actions, reasoning and required human approval. This deterministic demonstration does not call Gemini or save a plan to the database.

Explanation received: see Appendix J. Personal comparison evidence is in Appendix I. Review the reconciled summary and prepare for the viva.

Appendix F.2 - Member 2 technical execution

M.Thanusikan | IT24103609
Resource and Inventory Management and SituationAnalysisAgent

Historical technical execution - run by Codex; later personal evidence and explanations are recorded separately.
Operator: OpenAI Codex. Started 06 October 2026, 22:56:49 (Asia/Colombo, UTC+05:30). This historical record documents component tests and actual agent behaviour. Later submitted personal runs appear in Appendix G; written explanations are now recorded in Member_Explanations.md; viva assessment remains separate.

Runner: run-member2-tests.ps1. Exact commands, operator and timestamps are retained in execution.json.

Test area	Passed	Failed	Skipped
ResourcesControllerTests	3	0	0
InventoryControllerTests	3	0	0
SheltersControllerTests	3	0	0
GeolocationServiceTests	2	0	0
SituationAnalysisAgentTests	1	0	0
Python agent tests	5	0	0
Total selected existing tests	17	0	0
Separate agent demonstration scenarios	2	0	0

M2-D01 - Flood with inventory shortages

Input: Flood emergency; priority High; inventory Medical=20, Food=100; resources 1; shelters 1: 10/10 occupied, 2: 30/100 occupied.

Observed: urgency_score = 85; nearest_shelter.id = 2; resource_deficiencies.0.isDeficit = true; llm_enhanced = false
Scenario result: PASS

M2-D02 - Adequate stock but no shelter capacity

Input: Routine assistance; priority Low; inventory Medical=50, Food=100; resources 3; shelters 1: 10/10 occupied.

Observed: urgency_score = 25; nearest_shelter = null; resource_deficiencies.0.isDeficit = false; resource_deficiencies.2.isDeficit = false
Scenario result: PASS

Evidence folder: members/member2-IT24103609/results/ai-assisted-20261006-225649-753197
Open results.html for named test outcomes and agent-demo.html for full inputs and actual outputs. Raw backend.trx, agents.xml, logs, source hashes and JSON records are included.

Scope. These are reruns of existing cases, not additional unique tests; group totals remain 270 automated cases and 272 inventory records. Selected backend fixtures use InMemory/TestServer and agent demos use the real Python classes in deterministic mode, without live LLM calls.

Incident priority, inventory, resources and shelters produce urgency and shortage analysis. The current method named nearest_shelter selects the eligible shelter with most available capacity; it does not calculate geographic distance. Backend tests here use InMemory, not PostgreSQL.

Explanation received: see Appendix J. Personal comparison evidence is in Appendix I. Review the reconciled summary and prepare for the viva.

Appendix F.3 - Member 3 technical execution

Denesh N | IT24102826
Dispatch and Task Assignment and ResourceAllocationAgent

Historical technical execution - run by Codex; later personal evidence and explanations are recorded separately.
Operator: OpenAI Codex. Started 06 October 2026, 22:56:54 (Asia/Colombo, UTC+05:30). This historical record documents component tests and actual agent behaviour. Later submitted personal runs appear in Appendix G; written explanations are now recorded in Member_Explanations.md; viva assessment remains separate.

Runner: run-member3-tests.ps1. Exact commands, operator and timestamps are retained in execution.json.

Test area	Passed	Failed	Skipped
DispatchAllocationTests	7	0	0
DispatchesControllerTests	6	0	0
DispatchValidationServiceTests	10	0	0
Python agent tests	3	0	0
Total selected existing tests	26	0	0
Separate agent demonstration scenarios	2	0	0

M3-D01 - Covered dispatch requirement

Input: Incident 7; priority Medium; requirement 1 Vehicle; available resources 1; volunteers 1.

Observed: resource_gaps = []; allocated_resources = [{"id": 1, "name": "Response van", "resourceType": "Vehicle", "status": "Available", "purpose": "Transport"}]; allocation_plan.requires_human_approval = true; allocation_plan.recommendation = APPROVE
Scenario result: PASS

M3-D02 - Insufficient resources for high-priority dispatch

Input: Incident 8; priority High; requirement default requirements for priority; available resources 0; volunteers 0.

Observed: resource_gaps = [{"resource_type": "Vehicle", "missing_qty": 2, "message": "Need 2 more Vehicle resources to meet the requirement."}, {"resource_type": "Medical", "missing_qty": 2, "message": "Need 2 more Medical resources to meet the requirement."}, {"resource_type": "Personnel", "missing_qty": 2, "message": "Need 2 more Personnel resources to meet the requirement."}]; allocated_resources = []; allocation_plan.requires_human_approval = true; allocation_plan.recommendation = REVISE
Scenario result: PASS

Evidence folder: members/member3-IT24102826/results/ai-assisted-20261006-225654-881738
Open results.html for named test outcomes and agent-demo.html for full inputs and actual outputs. Raw backend.trx, agents.xml, logs, source hashes and JSON records are included.

Scope. These are reruns of existing cases, not additional unique tests; group totals remain 270 automated cases and 272 inventory records. Selected backend fixtures use InMemory/TestServer and agent demos use the real Python classes in deterministic mode, without live LLM calls.

A dispatch requirement, resource pool and available volunteers produce a proposed allocation and shortage list. Human approval is always required; the displayed proposal does not execute a dispatch.

Explanation received: see Appendix J. Personal comparison evidence is in Appendix I. Review the reconciled summary and prepare for the viva.

Appendix G - Member personal testing screenshots

Six terminal screenshots supplied on 7 October 2026 are matched to the existing dated runs for members 1 to 3. The commands declare each named operator and personal execution. Raw TRX/XML reports and agent JSON outputs confirm the recorded results. The original PNGs are preserved unchanged.

Member	Backend	Agent	Total tests	Demos
Jathurshy.S	9	2	11	2
M.Thanusikan	12	5	17	2
Denesh N	23	3	26	2

These are 54 passing existing tests and six separately checked demonstrations, not additional unique group cases. Group totals remain 270 automated cases and 272 inventory records. No selected tests failed or were skipped.

The screenshots show terminal execution. Generated results.html and agent-demo.html pages and their raw inputs/outputs are included in each run folder; browser screenshots of those pages were not supplied. Personal changed-input screenshots are now included in Appendix I; all four explanations are now recorded in Member_Explanations.md.

Jathurshy.S | IT24103355
Started: 2026-10-07T16:24:59.058648+05:30
Evidence: members/member1-IT24103355/results/personal-20261007-162459-058365

M.Thanusikan | IT24103609
Started: 2026-10-07T16:26:42.902996+05:30
Evidence: members/member2-IT24103609/results/personal-20261007-162642-902561

Denesh N | IT24102826
Started: 2026-10-07T16:27:57.690871+05:30
Evidence: members/member3-IT24102826/results/personal-20261007-162757-690323

Personal_Execution_Evidence.md links each report and screenshot gallery. members/personal-evidence.json retains screenshot and raw-report SHA-256 hashes.

Appendix G.1.1 - Backend command and passing cases

Jathurshy.S | IT24103355

Declared operator: Jathurshy.S; started 2026-10-07T16:24:59.058648+05:30

```
PS D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\members\member1-IT24103355> .\run-member1-tests.ps1 -ExecutionKind personal -
Operator 'Jathurshy.S'
Running backend: dotnet test "D:\Sliit Subjects\Y3 S1\SEF\Assignment\SEF\emergency-resource-coordination-platform\backend-t
ests\ERCP.Api.Tests.csproj" --filter FullyQualifiedName~ERCP.Api.Tests.IncidentControllerTests.|FullyQualifiedName~ERCP.Api
.Tests.IncidentPlannerServiceTests. --logger trx;LogFileName=backend.trx --logger console;verbosity=normal --results-direct
ory "D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\members\member1-IT24103355\results\personal-20261007-162459-058365"
  Determining projects to restore...
  All projects are up-to-date for restore.
  ERCP.Api -> D:\Sliit Subjects\Y3 S1\SEF\Assignment\SEF\emergency-resource-coordination-platform\backend\bin\Debug\net8.0\
ERCP.Api.dll
  ERCP.Api.Tests -> D:\Sliit Subjects\Y3 S1\SEF\Assignment\SEF\emergency-resource-coordination-platform\backend-tests\bin\D
ebug\net8.0\ERCP.Api.Tests.dll
Test run for D:\Sliit Subjects\Y3 S1\SEF\Assignment\SEF\emergency-resource-coordination-platform\backend-tests\bin\Debug\ne
t8.0\ERCP.Api.Tests.dll (.NETCoreApp,Version=v8.0)
A total of 1 test files matched the specified pattern.
[xUnit.net 00:00:00.00] xUnit.net VSTest Adapter v2.5.3.1+6b60a9e56a (64-bit .NET 8.0.29)
[xUnit.net 00:00:00.78]   Discovering: ERCP.Api.Tests
[xUnit.net 00:00:00.96]   Discovered:  ERCP.Api.Tests
[xUnit.net 00:00:00.96]   Starting:    ERCP.Api.Tests
[xUnit.net 00:00:02.17]   Finished:    ERCP.Api.Tests
  Passed ERCP.Api.Tests.IncidentControllerTests.CreateIncident_Successfully_ReturnsCreatedAndLogsTimelineEvent [928 ms]
  Passed ERCP.Api.Tests.IncidentPlannerServiceTests.GeneratePlan_WithAvailableResourcesAndVolunteers_AgentReceivesData [1 s
]
  Passed ERCP.Api.Tests.IncidentControllerTests.GeneratePlan_CreatesAndReturnsIncidentPlan_AndLogsTimelineEvent [191 ms]
  Passed ERCP.Api.Tests.IncidentControllerTests.GetTimeline_ReturnsEventsInChronologicalOrder [15 ms]
  Passed ERCP.Api.Tests.IncidentControllerTests.UpdatePriority_CreatesPriorityChangedTimelineEvent [8 ms]
  Passed ERCP.Api.Tests.IncidentControllerTests.GetIncident_NonExistingId_ReturnsNotFound [12 ms]
  Passed ERCP.Api.Tests.IncidentControllerTests.UpdateStatus_InvalidTransition_ReportedToResolved_ReturnsBadRequest [1 ms]
  Passed ERCP.Api.Tests.IncidentControllerTests.UpdateStatus_ValidTransition_ReportedToVerified_Succeeds [5 ms]
  Passed ERCP.Api.Tests.IncidentPlannerServiceTests.GeneratePlan_WithNoResources_AndNoVolunteers_ReportsNoAvailable [91 ms]
Results File: D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\members\member1-IT24103355\results\personal-20261007-162459-058365
\backend.trx
```

Raw reports confirm 9 backend and 2 agent tests passed, with no failures or skips; 2 separate demonstrations passed.

Original PNG: members/member1-IT24103355/results/personal-20261007-162459-058365/01-backend-tests.png
The image is unchanged and scaled proportionally.

Appendix G.1.2 - Agent tests and completed run summary

Jathurshy.S | IT24103355

Declared operator: Jathurshy.S; started 2026-10-07T16:24:59.058648+05:30

```
Test Run Successful.
Total tests: 9
     Passed: 9
 Total time: 4.5140 Seconds
Running agents: C:\Users\USER\AppData\Local\Programs\Python\Python313\python.exe -m pytest "D:\Sliit Subjects\Y3 S1\SEF\Ass
ignment\QMS\scripts\test_agents.py::test_AI01_unapproved_tool_is_rejected[CoordinatorPlannerAgent]" "D:\Sliit Subjects\Y3 S
1\SEF\Assignment\QMS\scripts\test_agents.py::test_AI02_coordinator_output_requires_human_approval" -v "--junitxml=D:\Sliit
Subjects\Y3 S1\SEF\Assignment\QMS\members\member1-IT24103355\results\personal-20261007-162459-058365\agents.xml"
============================= test session starts =============================
platform win32 -- Python 3.13.2, pytest-8.3.3, pluggy-1.6.0 -- C:\Users\USER\AppData\Local\Programs\Python\Python313\python
.exe
cachedir: .pytest_cache
rootdir: D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\scripts
configfile: pytest.ini
plugins: anyio-4.8.0, asyncio-0.24.0
asyncio: mode=Mode.STRICT, default_loop_scope=function
collecting ... collected 2 items

..\..\QMS\scripts\test_agents.py::test_AI01_unapproved_tool_is_rejected[CoordinatorPlannerAgent] PASSED [ 50%]
..\..\QMS\scripts\test_agents.py::test_AI02_coordinator_output_requires_human_approval PASSED [100%]

- generated xml file: D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\members\member1-IT24103355\results\personal-20261007-16245
9-058365\agents.xml -
============================== 2 passed in 0.15s ==============================
Member 1 agent scenarios: 2/2 passed. D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\members\member1-IT24103355\results\persona
l-20261007-162459-058365\agent-demo.html
PASS: D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\members\member1-IT24103355\results\personal-20261007-162459-058365\results
.html
```

Raw reports confirm 9 backend and 2 agent tests passed, with no failures or skips; 2 separate demonstrations passed.

Original PNG: members/member1-IT24103355/results/personal-20261007-162459-058365/02-agent-tests-and-summary.png
The image is unchanged and scaled proportionally.

Appendix G.2.1 - Backend command and passing cases

M.Thanusikan | IT24103609

Declared operator: M.Thanusikan; started 2026-10-07T16:26:42.902996+05:30

```
PS D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\members\member2-IT24103609> .\run-member2-tests.ps1 -ExecutionKind personal -
Operator 'M.Thanusikan'
Running backend: dotnet test "D:\Sliit Subjects\Y3 S1\SEF\Assignment\SEF\emergency-resource-coordination-platform\backend-t
ests\ERCP.Api.Tests.csproj" --filter FullyQualifiedName~ERCP.Api.Tests.ResourcesControllerTests.|FullyQualifiedName~ERCP.Ap
i.Tests.InventoryControllerTests.|FullyQualifiedName~ERCP.Api.Tests.SheltersControllerTests.|FullyQualifiedName~ERCP.Api.Te
sts.GeolocationServiceTests.|FullyQualifiedName~ERCP.Api.Tests.SituationAnalysisAgentTests. --logger trx;LogFileName=backen
d.trx --logger console;verbosity=normal --results-directory "D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\members\member2-IT2
4103609\results\personal-20261007-162642-902561"
  Determining projects to restore...
  All projects are up-to-date for restore.
  ERCP.Api -> D:\Sliit Subjects\Y3 S1\SEF\Assignment\SEF\emergency-resource-coordination-platform\backend\bin\Debug\net8.0\
ERCP.Api.dll
  ERCP.Api.Tests -> D:\Sliit Subjects\Y3 S1\SEF\Assignment\SEF\emergency-resource-coordination-platform\backend-tests\bin\D
ebug\net8.0\ERCP.Api.Tests.dll
Test run for D:\Sliit Subjects\Y3 S1\SEF\Assignment\SEF\emergency-resource-coordination-platform\backend-tests\bin\Debug\ne
t8.0\ERCP.Api.Tests.dll (.NETCoreApp,Version=v8.0)
A total of 1 test files matched the specified pattern.
[xUnit.net 00:00:00.00] xUnit.net VSTest Adapter v2.5.3.1+6b60a9e56a (64-bit .NET 8.0.29)
[xUnit.net 00:00:00.07]   Discovering: ERCP.Api.Tests
[xUnit.net 00:00:00.11]   Discovered:  ERCP.Api.Tests
[xUnit.net 00:00:00.12]   Starting:    ERCP.Api.Tests
  Passed ERCP.Api.Tests.GeolocationServiceTests.CalculateHaversineDistanceKm_ReturnsAccurateKmDistance [3 ms]
  Passed ERCP.Api.Tests.InventoryControllerTests.AdjustStock_SubtractsQuantity_UpdatesStatus [415 ms]
  Passed ERCP.Api.Tests.SheltersControllerTests.UpdateOccupancy_UpdatesStatusToFull_WhenCapacityReached [415 ms]
  Passed ERCP.Api.Tests.InventoryControllerTests.AdjustStock_ReturnsBadRequest_WhenResultingQuantityIsNegative [23 ms]
  Passed ERCP.Api.Tests.SheltersControllerTests.CreateShelter_ReturnsCreated [25 ms]
[xUnit.net 00:00:00.66]   Finished:    ERCP.Api.Tests
  Passed ERCP.Api.Tests.SheltersControllerTests.UpdateOccupancy_ReturnsBadRequest_WhenOccupancyExceedsTotalCapacity [< 1 ms
]
  Passed ERCP.Api.Tests.InventoryControllerTests.CreateInventoryItem_SetsLowStock_WhenBelowThreshold [2 ms]
  Passed ERCP.Api.Tests.GeolocationServiceTests.GetNearestShelters_OrdersSheltersByProximity [462 ms]
  Passed ERCP.Api.Tests.ResourcesControllerTests.GetAvailable_ReturnsOnlyAvailableResources [485 ms]
  Passed ERCP.Api.Tests.ResourcesControllerTests.UpdateStatus_ChangesResourceStatus [2 ms]
  Passed ERCP.Api.Tests.ResourcesControllerTests.Create_AddsResource_ReturnsCreated [1 ms]
```

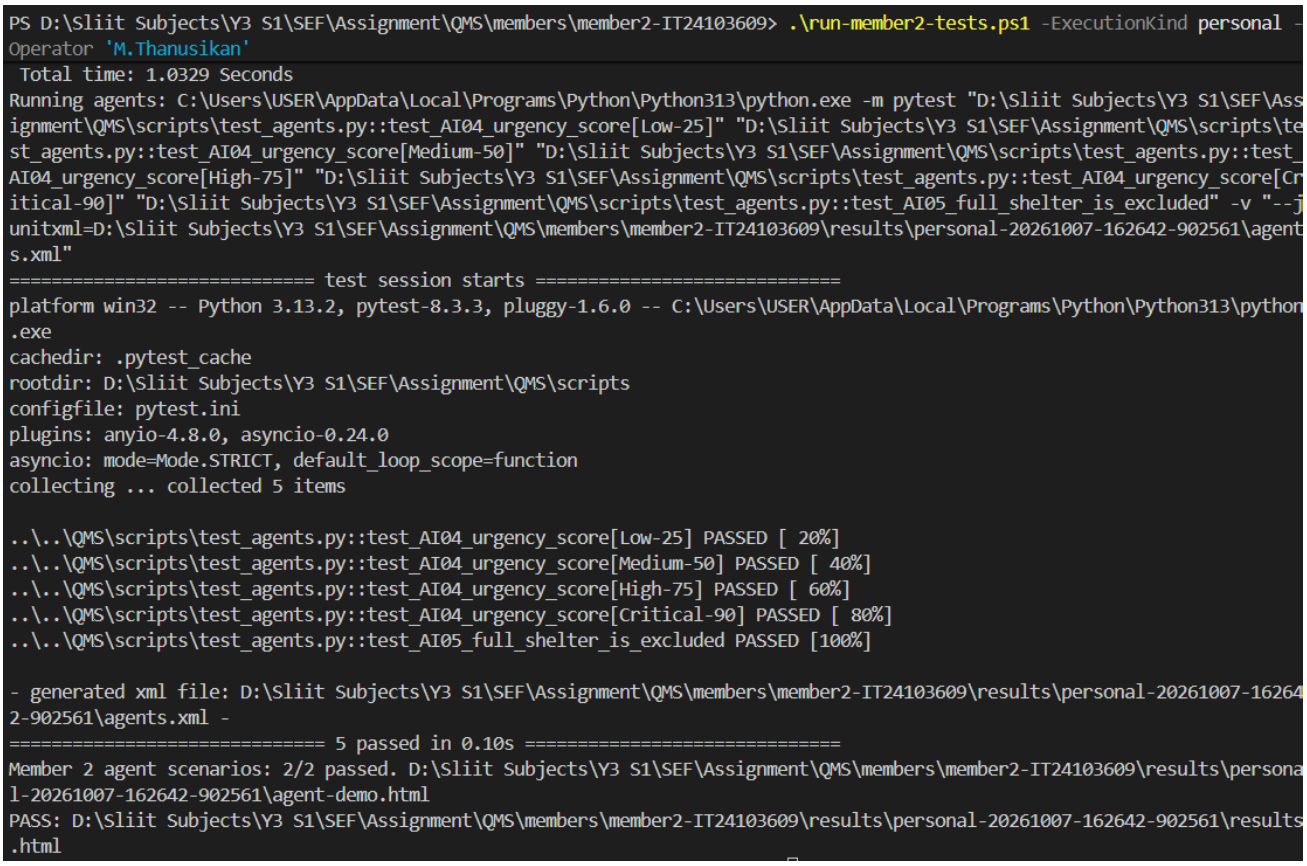
Raw reports confirm 12 backend and 5 agent tests passed, with no failures or skips; 2 separate demonstrations passed.

Original PNG: members/member2-IT24103609/results/personal-20261007-162642-902561/01-backend-tests.png
The image is unchanged and scaled proportionally.

Appendix G.2.2 - Agent tests and completed run summary

M.Thanusikan | IT24103609

Declared operator: M.Thanusikan; started 2026-10-07T16:26:42.902996+05:30



Raw reports confirm 12 backend and 5 agent tests passed, with no failures or skips; 2 separate demonstrations passed.

Original PNG: members/member2-IT24103609/results/personal-20261007-162642-902561/02-agent-tests-and-summary.png
The image is unchanged and scaled proportionally.

Appendix G.3.1 - Backend command and passing cases

Denesh N | IT24102826

Declared operator: Denesh N; started 2026-10-07T16:27:57.690871+05:30

```
PS D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\members\member3-IT24102826> .\run-member3-tests.ps1 -ExecutionKind personal -
Operator 'Denesh N'
Running backend: dotnet test "D:\Sliit Subjects\Y3 S1\SEF\Assignment\SEF\emergency-resource-coordination-platform\backend-t
ests\ERCP.Api.Tests.csproj" --filter FullyQualifiedName~ERCP.Api.Tests.DispatchAllocationTests.|FullyQualifiedName~ERCP.Api
.Tests.DispatchesControllerTests.|FullyQualifiedName~ERCP.Api.Tests.DispatchValidationServiceTests. --logger trx;LogFileNam
e=backend.trx --logger console;verbosity=normal --results-directory "D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\members\mem
ber3-IT24102826\results\personal-20261007-162757-690323"
  Determining projects to restore...
  All projects are up-to-date for restore.
  ERCP.Api -> D:\Sliit Subjects\Y3 S1\SEF\Assignment\SEF\emergency-resource-coordination-platform\backend\bin\Debug\net8.0\
ERCP.Api.dll
  ERCP.Api.Tests -> D:\Sliit Subjects\Y3 S1\SEF\Assignment\SEF\emergency-resource-coordination-platform\backend-tests\bin\D
ebug\net8.0\ERCP.Api.Tests.dll
Test run for D:\Sliit Subjects\Y3 S1\SEF\Assignment\SEF\emergency-resource-coordination-platform\backend-tests\bin\Debug\ne
t8.0\ERCP.Api.Tests.dll (.NETCoreApp,Version=v8.0)
A total of 1 test files matched the specified pattern.
[xUnit.net 00:00:00.00] xUnit.net VSTest Adapter v2.5.3.1+6b60a9e56a (64-bit .NET 8.0.29)
[xUnit.net 00:00:00.07]   Discovering: ERCP.Api.Tests
[xUnit.net 00:00:00.11]   Discovered:  ERCP.Api.Tests
[xUnit.net 00:00:00.12]   Starting:    ERCP.Api.Tests
  Passed ERCP.Api.Tests.DispatchValidationServiceTests.ValidateDispatch_UnknownDispatch_ThrowsNotFound [457 ms]
  Passed ERCP.Api.Tests.DispatchesControllerTests.UpdateStatus_ApprovalWithNewerFailedSafetyAudit_IsRejected [468 ms]
  Passed ERCP.Api.Tests.DispatchAllocationTests.ValidateDispatch_UnavailableResourceRejectsAllocation [493 ms]
  Passed ERCP.Api.Tests.DispatchValidationServiceTests.ValidateDispatch_UnavailableLinkedResource_ReturnsReject_WithR1 [39
ms]
  Passed ERCP.Api.Tests.DispatchValidationServiceTests.ValidateDispatch_ShelterOverAllocation_ReturnsRevise_WithS2 [25 ms]
  Passed ERCP.Api.Tests.DispatchValidationServiceTests.ValidateDispatch_UnavailableVolunteer_ReturnsReject_WithV1 [1 ms]
  Passed ERCP.Api.Tests.DispatchesControllerTests.ProposeAllocation_AttachesIncidentPlanAndMovesToPendingApproval [64 ms]
  Passed ERCP.Api.Tests.DispatchesControllerTests.UpdateTask_RejectsChangesAfterApproval [2 ms]
  Passed ERCP.Api.Tests.DispatchValidationServiceTests.ValidateDispatch_InsufficientInventory_ReturnsRevise_WithI1 [26 ms]
  Passed ERCP.Api.Tests.DispatchesControllerTests.UpdateTask_ChangesAssignmentAndRefreshesDispatchTimestamp [19 ms]
  Passed ERCP.Api.Tests.DispatchesControllerTests.UpdateStatus_ApprovalWithoutSafetyAudit_IsRejected [1 ms]
  Passed ERCP.Api.Tests.DispatchesControllerTests.UpdateStatus_ApprovalWithLatestPassingSafetyAudit_Succeeds [1 ms]
```

Raw reports confirm 23 backend and 3 agent tests passed, with no failures or skips; 2 separate demonstrations passed.

Original PNG: members/member3-IT24102826/results/personal-20261007-162757-690323/01-backend-tests.png
The image is unchanged and scaled proportionally.

Appendix G.3.2 - Agent tests and completed run summary

Denesh N | IT24102826

Declared operator: Denesh N; started 2026-10-07T16:27:57.690871+05:30

```
PS D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\members\member3-IT24102826> .\run-member3-tests.ps1 -ExecutionKind personal -
  Passed ERCP.Api.Tests.DispatchValidationServiceTests.ValidateDispatch_AvailableVolunteer_ReturnsApprove [2 ms]
  Passed ERCP.Api.Tests.DispatchValidationServiceTests.ValidateDispatch_NoTasks_ReturnsRevise_WithW0 [2 ms]
Results File: D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\members\member3-IT24102826\results\personal-20261007-162757-690323
\backend.trx

Test Run Successful.
Total tests: 23
     Passed: 23
 Total time: 1.2258 Seconds
Running agents: C:\Users\USER\AppData\Local\Programs\Python\Python313\python.exe -m pytest "D:\Sliit Subjects\Y3 S1\SEF\Ass
ignment\QMS\scripts\test_agents.py::test_AI01_unapproved_tool_is_rejected[ResourceAllocationAgent]" "D:\Sliit Subjects\Y3 S
1\SEF\Assignment\QMS\scripts\test_agents.py::test_AI03_allocation_shortage_is_explicit" "D:\Sliit Subjects\Y3 S1\SEF\Assign
ment\QMS\scripts\test_agents.py::test_AI11_injected_text_does_not_bypass_deterministic_approval" -v "--junitxml=D:\Sliit Su
bjects\Y3 S1\SEF\Assignment\QMS\members\member3-IT24102826\results\personal-20261007-162757-690323\agents.xml"
============================= test session starts =============================
platform win32 -- Python 3.13.2, pytest-8.3.3, pluggy-1.6.0 -- C:\Users\USER\AppData\Local\Programs\Python\Python313\python
.exe
cachedir: .pytest_cache
rootdir: D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\scripts
configfile: pytest.ini
plugins: anyio-4.8.0, asyncio-0.24.0
asyncio: mode=Mode.STRICT, default_loop_scope=function
collecting ... collected 3 items

..\..\QMS\scripts\test_agents.py::test_AI01_unapproved_tool_is_rejected[ResourceAllocationAgent] PASSED [ 33%]
..\..\QMS\scripts\test_agents.py::test_AI03_allocation_shortage_is_explicit PASSED [ 66%]
..\..\QMS\scripts\test_agents.py::test_AI11_injected_text_does_not_bypass_deterministic_approval PASSED [100%]

- generated xml file: D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\members\member3-IT24102826\results\personal-20261007-16275
7-690323\agents.xml -
============================== 3 passed in 0.10s ==============================
Member 3 agent scenarios: 2/2 passed. D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\members\member3-IT24102826\results\persona
l-20261007-162757-690323\agent-demo.html
```

Raw reports confirm 23 backend and 3 agent tests passed, with no failures or skips; 2 separate demonstrations passed.

Original PNG: members/member3-IT24102826/results/personal-20261007-162757-690323/02-agent-tests-and-summary.png
The image is unchanged and scaled proportionally.

Appendix H - Changed-input comparisons

On 8 October 2026, OpenAI Codex ran two scenarios per member against the actual Python agents. All eight scenarios passed. These are separate technical demonstrations of existing behaviour; group totals remain 270 automated tests and 272 inventory records.

The exact command, prepared prediction saved before execution, full JSON inputs, observed outputs and agent source hash are retained for each run. Later supplied personal terminal screenshots are retained in Appendix I, with matching dated records. These are separate from the technical runs below.

Deneshkar submitted written explanation

Received in the conversation on 8 October 2026 and reproduced verbatim below. This is a written explanation, not confirmation of a new personal execution or completion of the viva.

The 10:00–11:00 shift passes because the existing shift finishes exactly at 10:00. The new shift starts at 10:00, so the two shifts only touch at the boundary; their working times do not overlap. Therefore, the agent returns APPROVE.

When I change the new shift's start time to 09:30, it overlaps the existing 09:00–10:00 shift for 30 minutes, from 09:30 to 10:00. The agent detects this as double booking. Because the problem can be fixed by changing the shift time, it returns REVISE instead of APPROVE.

Source: members/member4-IT24103828/Understanding_Submitted_20261008.md

All four explanations are now recorded in Member_Explanations.md. All four supplied personal comparison execution screenshots, retained in Appendix I. Changed_Input_Results.md links the recorded technical reports; each member folder contains Viva_Changed_Input_Guide.md and run-memberN-changed-input.ps1.

Appendix H.1 - Jathurshy.S

Incident Management + CoordinatorPlannerAgent
Operator: OpenAI Codex; 2026-10-08T00:31:56.222748+05:30

Command actually executed: C:\Users\USER\AppData\Local\Programs\Python\Python313\python.exe QMS/scripts/run_changed_input.py --member 1 --operator "OpenAI Codex" --execution-kind ai-assisted

Input change: resources: 1 to 0; volunteers: 1 to 0. The incident, High priority and affected count stay unchanged.

Expected before execution: With resources and volunteers removed, reasoning should report both shortages while requiresHumanApproval remains true.

Scenario	Expected check	Observed result
M1-C01	incidentId equals 7	7; PASS
M1-C01	requiresHumanApproval equals true	true; PASS
M1-C01	actions nonempty	2 item(s); PASS
M1-C02	incidentId equals 7	7; PASS
M1-C02	requiresHumanApproval equals true	true; PASS
M1-C02	reasoning contains "No available resources found"	Contains "No available resources found"; PASS
M1-C02	reasoning contains "No available volunteers found"	Contains "No available volunteers found"; PASS

Removing resources and volunteers causes the planner to report both shortages. Human approval remains true in both outputs. The plan is a proposal, not an independently executed emergency response.

Saved evidence: members/member1-IT24103355/results/changed-20261008-003156-222748 comparison.html, comparison.json, execution.log, prediction-before-run.json and agent-demo.html. Full input/output is preserved; the table summarizes the observed checks.

Result: 2/2 scenarios PASS. This is the earlier Codex-operated technical run. The later supplied personal run appears in Appendix I.

Appendix H.2 - M.Thanusikan

Resource and Inventory Management + SituationAnalysisAgent
Operator: OpenAI Codex; 2026-10-08T00:31:56.384058+05:30

Command actually executed: C:\Users\USER\AppData\Local\Programs\Python\Python313\python.exe QMS/scripts/run_changed_input.py --member 2 --operator "OpenAI Codex" --execution-kind ai-assisted

Input change: Medical stock: 20 to 50; available vehicles: 1 to 3; shelter 2 occupancy: 30/100 to 100/100. Food stays 100; shelter 1 stays 10/10; incident and High priority unchanged.

Expected before execution: Medical and vehicle deficits should clear; no shelter should be selected because both are full. Urgency remains 85 because the incident is unchanged.

Scenario	Expected check	Observed result
M2-C01	urgency_score equals 85	85; PASS
M2-C01	nearest_shelter.id equals 2	2; PASS
M2-C01	resource_deficiencies.0.isDeficit equals true	true; PASS
M2-C01	llm_enhanced equals false	false; PASS
M2-C02	urgency_score equals 85	85; PASS
M2-C02	nearest_shelter equals null	null; PASS
M2-C02	resource_deficiencies.0.isDeficit equals false	false; PASS
M2-C02	resource_deficiencies.1.isDeficit equals false	false; PASS
M2-C02	resource_deficiencies.2.isDeficit equals false	false; PASS
M2-C02	llm_enhanced equals false	false; PASS

Medical stock of 50, food stock of 100 and three available vehicles meet the current fixed baselines. With both shelters full, nearest_shelter is null. Despite its name, this field selects the active shelter with greatest available capacity, not geographic distance. Urgency stays 85 because the High-priority flood incident did not change.

Saved evidence: members/member2-IT24103609/results/changed-20261008-003156-384058 comparison.html, comparison.json, execution.log, prediction-before-run.json and agent-demo.html. Full input/output is preserved; the table summarizes the observed checks.

Result: 2/2 scenarios PASS. This is the earlier Codex-operated technical run. The later supplied personal run appears in Appendix I.

Appendix H.3 - Denesh N

Dispatch and Task Assignment + ResourceAllocationAgent
Operator: OpenAI Codex; 2026-10-08T00:31:56.602167+05:30

Command actually executed: C:\Users\USER\AppData\Local\Programs\Python\Python313\python.exe QMS/scripts/run_changed_input.py --member 3 --operator "OpenAI Codex" --execution-kind ai-assisted

Input change: Priority: Medium to High; resources: 1 to 0; volunteers: 1 to 0. The explicit one-vehicle requirement and incident ID stay unchanged.

Expected before execution: The missing vehicle should produce a resource gap and REVISE; human approval remains required.

Scenario	Expected check	Observed result
M3-C01	resource_gaps empty	0 item(s); PASS
M3-C01	allocated_resources nonempty	1 item(s); PASS
M3-C01	allocation_plan.requires_human_approval equals true	true; PASS
M3-C01	allocation_plan.recommendation equals "APPROVE"	"APPROVE"; PASS
M3-C02	resource_gaps nonempty	3 item(s); PASS
M3-C02	allocated_resources empty	0 item(s); PASS
M3-C02	allocation_plan.requires_human_approval equals true	true; PASS
M3-C02	allocation_plan.recommendation equals "REVISE"	"REVISE"; PASS

The Medium case allocates the available vehicle and has no gaps. The High case uses the current priority-based estimate, replacing the explicit one-vehicle input requirement with two vehicles, two medical resources and two personnel resources. With no resources, each category has a gap of two and the recommendation is REVISE. Human approval remains true. This comparison changes priority and availability together; it does not isolate the effect of each change.

Saved evidence: members/member3-IT24102826/results/changed-20261008-003156-602167 comparison.html, comparison.json, execution.log, prediction-before-run.json and agent-demo.html. Full input/output is preserved; the table summarizes the observed checks.

Result: 2/2 scenarios PASS. This is the earlier Codex-operated technical run. The later supplied personal run appears in Appendix I.

Appendix H.4 - Deneshkar. P

Volunteer/Responder Management + ValidationSafetyAgent
Operator: OpenAI Codex; 2026-10-08T00:31:56.747636+05:30

Command actually executed: C:\Users\USER\AppData\Local\Programs\Python\Python313\python.exe QMS/scripts/run_changed_input.py --member 4 --operator "OpenAI Codex" --execution-kind ai-assisted

Input change: Only proposed_shift.start changes: 10:00 to 09:30 UTC. Existing shift remains 09:00-10:00, end remains 11:00, volunteer remains available.

Expected before execution: The original adjacent 10:00-11:00 shift should be valid and APPROVE. Changing only its start to 09:30 should produce is_valid=false, REVISE and V2_DOUBLE_BOOKING.

Scenario	Expected check	Observed result
M4-C01	is_valid equals true	true; PASS
M4-C01	recommendation equals "APPROVE"	"APPROVE"; PASS
M4-C01	checks.1.passed equals true	true; PASS
M4-C02	is_valid equals false	false; PASS
M4-C02	recommendation equals "REVISE"	"REVISE"; PASS
M4-C02	violations.0.code equals "V2_DOUBLE_BOOKING"	"V2_DOUBLE_BOOKING"; PASS

The overlap predicate uses half-open intervals: new_start < existing_end and existing_start < new_end. At 10:00 the first comparison is false; at 09:30 both are true. V2_DOUBLE_BOOKING is in the agent's revisable violation set, so this available volunteer receives REVISE. Availability alone is not sufficient for approval. APPROVE is only a recommendation; coordinator approval is a separate API operation.

Saved evidence: members/member4-IT24103828/results/changed-20261008-003156-747636 comparison.html, comparison.json, execution.log, prediction-before-run.json and agent-demo.html. Full input/output is preserved; the table summarizes the observed checks.

Result: 2/2 scenarios PASS. This is the earlier Codex-operated technical run. The later supplied personal run appears in Appendix I.

Appendix I.1 - Personal changed-input evidence

Jathurshy.S | IT24103355
Declared operator: Jathurshy.S
Started: 2026-10-08T10:52:02.730954+05:30

The supplied terminal screenshot matches the saved personal run. Both original and changed scenarios passed. The command, input summaries and actual results are visible; full input/output and the transcript remain in the dated results folder.

```
PS D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\members\member1-IT24103355> .\run-member1-changed-input.ps1 -Exec
utionKind personal -Operator 'Jathurshy.S'
Member 1 agent scenarios: 2/2 passed. D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\members\member1-IT24103355\res
ults\changed-20261008-105202-730954\agent-demo.html
M1-C01 {"incident_id": 7, "priority": "High", "resources": 1, "volunteers": 1} PASS
  incidentId => 7
  requiresHumanApproval => true
  actions => ["Secure perimeter at Demo Sector", "Deploy allocated resources and volunteers"]
M1-C02 {"incident_id": 7, "priority": "High", "resources": 0, "volunteers": 0} PASS
  incidentId => 7
  requiresHumanApproval => true
  reasoning => "1. Observed High priority Fire incident with 5 affected.\n2. Gathered real data: No available r
esources found.\n3. Gathered real data: No available volunteers found.\n4. Identified constraint: Lack of requi
red vehicles.\n5. Identified constraint: Lack of required volunteers.\n6. Formulating safe dispatch plan. Requi
res human validation before execution."
  reasoning => "1. Observed High priority Fire incident with 5 affected.\n2. Gathered real data: No available r
esources found.\n3. Gathered real data: No available volunteers found.\n4. Identified constraint: Lack of requi
red vehicles.\n5. Identified constraint: Lack of required volunteers.\n6. Formulating safe dispatch plan. Requi
res human validation before execution."
Report: D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\members\member1-IT24103355\results\changed-20261008-105202-7
30954\comparison.html
```

Observed result: Removing resources and volunteers causes the planner to report both shortages. Human approval remains true in both outputs. The plan is a proposal, not an independently executed emergency response.

Written explanation: Received via group representative; evidence-aligned summary retained in Member_Explanations.md; scenario corrections documented. The runner's prepared prediction is supplied guidance, not evidence of an independently authored pre-run prediction. Viva understanding is assessed separately.

Evidence: members/member1-IT24103355/results/changed-20261008-105202-730954 personal-terminal-results.png, comparison.html, comparison.json, execution.log and Screenshot_Evidence.md. Original image preserved unchanged.

These personal reruns do not increase the 270 unique group tests or 272 inventory records. Personal_Changed_Input_Evidence.md links all four evidence sets.

Appendix I.2 - Personal changed-input evidence

M.Thanusikan | IT24103609
Declared operator: M.Thanusikan
Started: 2026-10-08T10:52:29.627673+05:30

The supplied terminal screenshot matches the saved personal run. Both original and changed scenarios passed. The command, input summaries and actual results are visible; full input/output and the transcript remain in the dated results folder.

```
PS D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\members\member2-IT24103609> .\run-member2-changed-input.ps1 -Exec
utionKind personal -Operator 'M.Thanusikan'
Member 2 agent scenarios: 2/2 passed. D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\members\member2-IT24103609\res
ults\changed-20261008-105229-627673\agent-demo.html
M2-C01 {"incident": {"id": 7, "title": "Flood emergency", "priority": "High"}, "inventory": [{"category": "Medi
cal", "quantity": 20}, {"category": "Food", "quantity": 100}], "vehicles": 1, "shelters": [{"id": 1, "occupancy
": "10/10"}, {"id": 2, "occupancy": "30/100"}]} PASS
  urgency_score => 85
  nearest_shelter.id => 2
  resource_deficiencies.0.isDeficit => true
  llm_enhanced => false
M2-C02 {"incident": {"id": 7, "title": "Flood emergency", "priority": "High"}, "inventory": [{"category": "Medi
cal", "quantity": 50}, {"category": "Food", "quantity": 100}], "vehicles": 3, "shelters": [{"id": 1, "occupancy
": "10/10"}, {"id": 2, "occupancy": "100/100"}]} PASS
  urgency_score => 85
  nearest_shelter => null
  resource_deficiencies.0.isDeficit => false
  resource_deficiencies.1.isDeficit => false
  resource_deficiencies.2.isDeficit => false
  llm_enhanced => false
Report: D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\members\member2-IT24103609\results\changed-20261008-105229-6
27673\comparison.html
```

Observed result: Medical stock of 50, food stock of 100 and three available vehicles meet the current fixed baselines. With both shelters full, nearest_shelter is null. Despite its name, this field selects the active shelter with greatest available capacity, not geographic distance. Urgency stays 85 because the High-priority flood incident did not change.

Written explanation: Received via group representative; evidence-aligned summary retained in Member_Explanations.md; scenario corrections documented. The runner's prepared prediction is supplied guidance, not evidence of an independently authored pre-run prediction. Viva understanding is assessed separately.

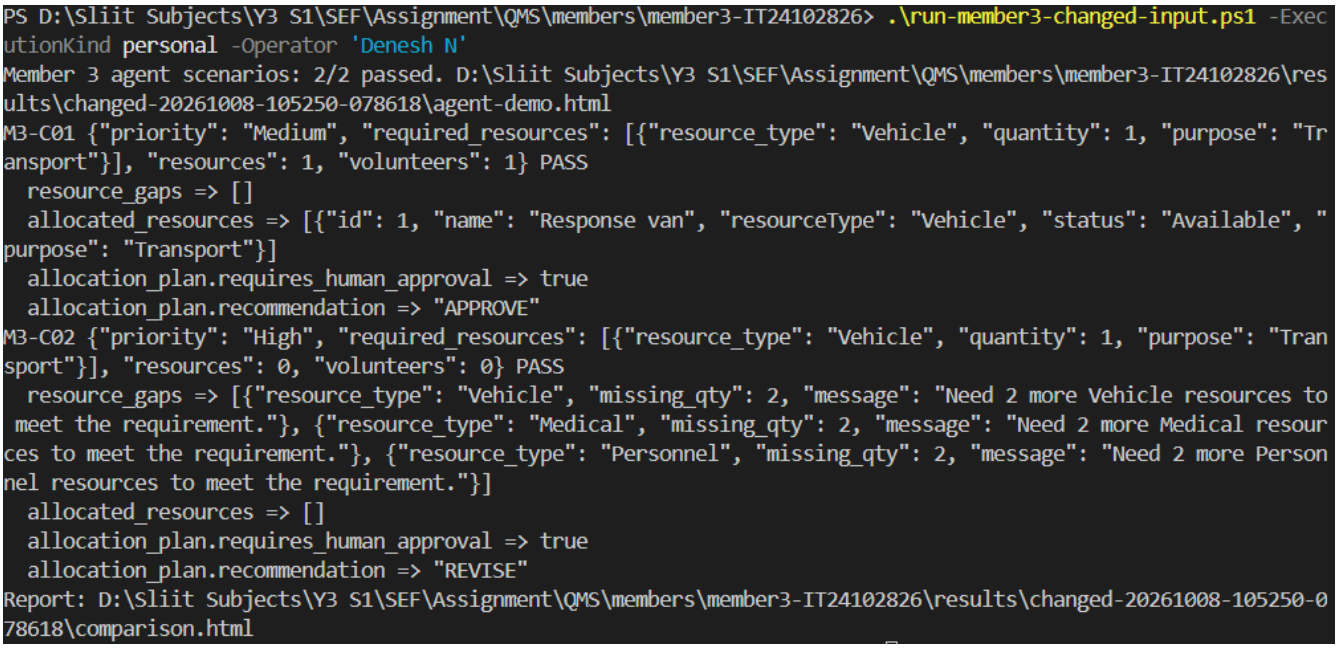
Evidence: members/member2-IT24103609/results/changed-20261008-105229-627673 personal-terminal-results.png, comparison.html, comparison.json, execution.log and Screenshot_Evidence.md. Original image preserved unchanged.

These personal reruns do not increase the 270 unique group tests or 272 inventory records. Personal_Changed_Input_Evidence.md links all four evidence sets.

Appendix I.3 - Personal changed-input evidence

Denesh N | IT24102826
Declared operator: Denesh N
Started: 2026-10-08T10:52:50.078618+05:30

The supplied terminal screenshot matches the saved personal run. Both original and changed scenarios passed. The command, input summaries and actual results are visible; full input/output and the transcript remain in the dated results folder.



Observed result: The Medium case allocates the available vehicle and has no gaps. The High case uses the current priority-based estimate, replacing the explicit one-vehicle input requirement with two vehicles, two medical resources and two personnel resources. With no resources, each category has a gap of two and the recommendation is REVISE. Human approval remains true. This comparison changes priority and availability together; it does not isolate the effect of each change.

Written explanation: Received via group representative; evidence-aligned summary retained in Member_Explanations.md. The runner's prepared prediction is supplied guidance, not evidence of an independently authored pre-run prediction. Viva understanding is assessed separately.

Evidence: members/member3-IT24102826/results/changed-20261008-105250-078618 personal-terminal-results.png, comparison.html, comparison.json, execution.log and Screenshot_Evidence.md. Original image preserved unchanged.

These personal reruns do not increase the 270 unique group tests or 272 inventory records. Personal_Changed_Input_Evidence.md links all four evidence sets.

Appendix I.4 - Personal changed-input evidence

Deneshkar. P | IT24103828
Declared operator: Deneshkar. P
Started: 2026-10-08T10:51:28.171178+05:30

The supplied terminal screenshot matches the saved personal run. Both original and changed scenarios passed. The command, input summaries and actual results are visible; full input/output and the transcript remain in the dated results folder.

```
PS D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\members\member4-IT24103828> .\run-member4-changed-input.ps1 -Exec
utionKind personal -Operator 'Deneshkar. P'
Member 4 agent scenarios: 2/2 passed. D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\members\member4-IT24103828\res
ults\changed-20261008-105128-171178\agent-demo.html
M4-C01 {"is_available": true, "existing_shift": "09:00-10:00 UTC", "proposed_shift": "10:00-11:00 UTC"} PASS
  is_valid => true
  recommendation => "APPROVE"
  checks.1.passed => true
M4-C02 {"is_available": true, "existing_shift": "09:00-10:00 UTC", "proposed_shift": "09:30-11:00 UTC"} PASS
  is_valid => false
  recommendation => "REVISE"
  violations.0.code => "V2_DOUBLE_BOOKING"
Report: D:\Sliit Subjects\Y3 S1\SEF\Assignment\QMS\members\member4-IT24103828\results\changed-20261008-105128-1
71178\comparison.html
```

Observed result: The overlap predicate uses half-open intervals: new_start < existing_end and existing_start < new_end. At 10:00 the first comparison is false; at 09:30 both are true. V2_DOUBLE_BOOKING is in the agent's revisable violation set, so this available volunteer receives REVISE. Availability alone is not sufficient for approval. APPROVE is only a recommendation; coordinator approval is a separate API operation.

Written explanation: Received via group representative; evidence-aligned summary retained in Member_Explanations.md. The runner's prepared prediction is supplied guidance, not evidence of an independently authored pre-run prediction. Viva understanding is assessed separately.

Evidence: members/member4-IT24103828/results/changed-20261008-105128-171178 personal-terminal-results.png, comparison.html, comparison.json, execution.log and Screenshot_Evidence.md. Original image preserved unchanged.

These personal reruns do not increase the 270 unique group tests or 272 inventory records. Personal_Changed_Input_Evidence.md links all four evidence sets.

Appendix J.1 - Member 1 explanation

Jathurshy.S | IT24103355
Incident Management + CoordinatorPlannerAgent

Received via the group representative on 8 October 2026. This is an evidence-aligned editorial summary; the original wording is preserved separately. Assessed viva completion is not claimed.

Evidence reconciliation: The submitted High-to-Medium description matches the older M1-D01/M1-D02 examples. The supplied 8 October screenshot and raw personal run use M1-C01/M1-C02 and keep High priority. The summary below corrects that distinction; it is an edited evidence summary, not a verbatim member statement.

What input changed?

Available resources changed from one to zero and volunteers from one to zero. Priority remained High, with the same incident ID, title, category and affected count.

What was expected?

The supplied explanation expects the plan to identify resource and volunteer shortages while retaining human approval. That expectation matches the executed comparison.

What actually happened?

Both scenarios passed. The changed plan reports no available resources and no available volunteers, identifies the missing vehicles and volunteers as constraints, and returns requiresHumanApproval=true.

Which assertion or business rule explains the result?

The checks assert incidentId=7, requiresHumanApproval=true, and reasoning containing both shortage messages. The planner exposes constraints and proposes a response; it does not independently execute the plan.

Personal scenario IDs: M1-C01, M1-C02. Both passed. Personal screenshot and dated raw results: Appendix I.1.

Original statement: members/member1-IT24103355/Explanation_As_Submitted_20261008.md
Evidence-aligned summary: members/member1-IT24103355/Written_Explanation.md

Appendix J.2 - Member 2 explanation

M.Thanusikan | IT24103609
Resource and Inventory Management + SituationAnalysisAgent

Received via the group representative on 8 October 2026. This is an evidence-aligned editorial summary; the original wording is preserved separately. Assessed viva completion is not claimed.

Evidence reconciliation: The submitted routine-assistance description and 85-to-25 urgency change match the older M2-D01/M2-D02 examples. The supplied personal run uses M2-C01/M2-C02: both incidents are the same High-priority flood, both scores are 85, and two shelters remain. The summary below corrects the distinction; the member should review these corrections before the viva.

What input changed?

Medical stock increased from 20 to 50 and available vehicles from one to three. Food stayed at 100. Shelter 1 remained 10/10; shelter 2 changed from 30/100 to 100/100. The same High-priority flood incident remained in both inputs.

What was expected?

For this executed pair, the evidence-aligned expectation is that stock deficits clear and no shelter is selected, while urgency stays 85. The supplied expectation of lower urgency belongs to the older D-series pair, not this C-series run.

What actually happened?

Both scenarios passed. Urgency was 85 in both. The original output selected shelter 2 and showed a medical deficit. The changed output returned nearest_shelter=null and all three resource-deficiency flags false.

Which assertion or business rule explains the result?

The checks assert urgency_score=85, nearest_shelter=null, and false deficit flags after the change. Medical 50, food 100 and three vehicles meet the current baselines. Full shelters are excluded; eligible shelters are ranked by available capacity, not geographic distance.

Personal scenario IDs: M2-C01, M2-C02. Both passed. Personal screenshot and dated raw results: Appendix I.2.

Original statement: members/member2-IT24103609/Explanation_As_Submitted_20261008.md
Evidence-aligned summary: members/member2-IT24103609/Written_Explanation.md

Appendix J.3 - Member 3 explanation

Denesh N | IT24102826
Dispatch and Task Assignment + ResourceAllocationAgent

Received via the group representative on 8 October 2026. This is an evidence-aligned editorial summary; the original wording is preserved separately. Assessed viva completion is not claimed.

Evidence reconciliation: The supplied explanation matches the personal run. Its submitted table labels D-series examples; the screenshot and dated comparison use M3-C01/M3-C02. The summary adds the observed High-priority requirement rule for clarity.

What input changed?

Priority changed from Medium to High, with available resources and volunteers each changing from one to zero. The input retained its explicit one-vehicle requirement.

What was expected?

The supplied expectation is an empty allocation, reported shortages and REVISE, while human approval remains required.

What actually happened?

Both scenarios passed. The original output allocated a response van, returned no gaps and APPROVE. The changed output allocated nothing, reported gaps of two Vehicles, two Medical resources and two Personnel, and returned REVISE. Human approval remained true.

Which assertion or business rule explains the result?

The checks assert empty/nonempty allocation and gap lists, requires_human_approval=true, and APPROVE/REVISE. In the current implementation, High priority replaces the explicit input requirement with the priority-based estimate. This comparison changes both priority and availability; it does not isolate their individual effects.

Personal scenario IDs: M3-C01, M3-C02. Both passed. Personal screenshot and dated raw results: Appendix I.3.

Original statement: members/member3-IT24102826/Explanation_As_Submitted_20261008.md
Evidence-aligned summary: members/member3-IT24102826/Written_Explanation.md

Appendix J.4 - Member 4 explanation

Deneshkar. P | IT24103828
Volunteer/Responder Management + ValidationSafetyAgent

Received via the group representative on 8 October 2026. This is an evidence-aligned editorial summary; the original wording is preserved separately. Assessed viva completion is not claimed.

Evidence reconciliation: The prior supplied explanation is preserved verbatim in Understanding_Submitted_20261008.md. The following four-part summary combines it with the verified M4-C01/M4-C02 personal run. It is not a new verbatim statement.

What input changed?

Only the proposed shift start changed, from 10:00 to 09:30 UTC. The existing shift stayed 09:00-10:00, the proposed end stayed 11:00 and the volunteer remained available.

What was expected?

The previously supplied explanation identifies adjacent shifts as non-overlapping and the 09:30 start as a 30-minute conflict requiring revision.

What actually happened?

Both personal scenarios passed. The adjacent case returned is_valid=true and APPROVE. The changed case returned is_valid=false, REVISE and V2_DOUBLE_BOOKING.

Which assertion or business rule explains the result?

The checks assert those returned fields. Half-open intervals overlap when new_start < existing_end and existing_start < new_end. Equality at 10:00 is allowed. V2_DOUBLE_BOOKING is revisable; an agent recommendation does not itself approve the dispatch.

Personal scenario IDs: M4-C01, M4-C02. Both passed. Personal screenshot and dated raw results: Appendix I.4.

Original statement: members/member4-IT24103828/Understanding_Submitted_20261008.md
Evidence-aligned summary: members/member4-IT24103828/Written_Explanation.md